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The youth dividend: Political impacts of demographic transitions

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ABSTRACT

This paper revisits the political economy of transitions to democracy. We extend a canonical rational choice model of democratization by introducing demographic structure, arguing that it shapes the material incentives for political mobilization. Younger cohorts benefit from democratic change over a longer horizon and face lower costs of participation, as they are less economically and socially integrated. As a result, increases in the youth share—such as during the demographic transition—can create a “window of opportunity” for democratization by strengthening the threat of political unrest faced by autocratic elites. We test this prediction using country-year panel data with detailed information on population age structure. Fixed-effects regressions reveal a robust positive relationship between the youth share and the likelihood of democratic improvement. This effect is strongest for the youngest cohorts, in particular individuals aged 15–19. To address concerns about endogeneity, we implement an instrumental variable strategy based on lagged fertility, including variation from neighboring countries. The IV estimates are consistent with the baseline results and are interpreted with appropriate caution. We also show that the effect of the youth share is amplified during economic downturns, highlighting a complementarity between demographic and macroeconomic forces.

“The reason the younger generation [in Iran] is taking this kind of risk is because they feel they have nothing to lose, they have no hope for the future [without regime change].”

—Ali Vaez, International Crisis Group, in *The NY Times* on September 26, 2022

1. Introduction

Youthful populations are often viewed as sources of political instability and conflict. Youth cohorts, not yet well integrated into socioeconomic institutions, have been considered to be the segment of the population with the lowest opportunity costs of participating in organized violence and rebel movements such that the presence of a “youth bulge” may be destabilizing in liberal democracies (Weber, 2013, 2019). However, the aforementioned opportunity cost effect might also induce them to engage in pro-democracy movements in more autocratic settings. Moreover, young adults are also those who have the longest life horizons and, because of this, the highest incentives to invest in long-term institutional and political changes which they will enjoy for more time periods. We argue that a high youth share increases the probability of observing a democratic improvement in societies that have not fully democratized.

In this paper, we theoretically and empirically investigate the relationship between demographic structure and political change. We show that the youth share has a robust positive impact on the probability of democratic improvement. This finding complements the theory of political transitions developed by Acemoglu and Robinson (2001, 2006) (henceforth AR), in which political elites may be induced to make concessions, in the form of institutional changes, under a threat of revolution. In AR, economic downturns open a “window of opportunity” for democratization, as lower opportunity costs increase the likelihood of political mobilization. We highlight a complementary channel: demographic structure—and the youth share in particular—can also generate such windows of opportunity. Moreover, economic and demographic forces interact. The impact of economic downturns on the probability of democratic improvement is amplified when the youth share is high. With this result, we also revisit the modernization theory (Lipset, 1959). While we do not argue that economic development *per se* is the primary driver of democratization, we show that demographic transformations associated with development play a key role in shaping political transitions.

Our conceptual framework builds on AR’s canonical rational choice model of democratization. The population is divided into two groups: the elite—who control the coercive power of the state and capture

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a share of economic output—and the non-elite. Non-elite individuals decide whether to participate in a democratic movement by comparing the short-run costs of participation with the long-run benefits of democracy. We introduce a demographic dimension into this framework. Individuals live for three periods—*young*, *mature*, and *old*—and both the costs and benefits of political participation vary over the life cycle. In particular, the youngest cohorts are only partially integrated into economic and social structures. Economically, they face lower opportunity costs due to weaker labor market attachment. Socially, they are not yet fully embedded in family and long-term commitments, which reduces the perceived non-monetary costs of political unrest. In addition, younger individuals have a longer time horizon over which to benefit from institutional change. As a result, young cohorts are disproportionately more likely to participate in democratic movements. A larger youth share therefore increases the threat of revolution, which may induce elites to make political concessions. A recession—modeled as a temporary decline in wages—further reduces the opportunity cost of participation, thereby reinforcing the effect of a high youth share.

We test these theoretical predictions using a country-year panel covering the period 1950–2018, which provides detailed information on population age structure. Guided by the theoretical distinction between “young” and more integrated cohorts, we construct our baseline measure of the youth share using the proportion of individuals aged 15–19 in the adult population (aged 15 and above). Using fixed-effects regressions with a comprehensive set of controls, we find a robust empirical regularity: a higher youth share is associated with a greater probability of democratic improvement, measured by an increase in the Polity2 index. Beyond statistical significance, we show that age structure provides meaningful additional explanatory power. Adding the youth-share variable to a specification that already includes the economic determinants emphasized by AR increases the within- R^2 by approximately 7.5%, indicating that age structure provides explanatory power beyond traditional economic downturns. We then assess whether the effect is driven by the youngest cohorts, as suggested by the model. Estimating alternative specifications with different age group definitions, we find that it is strongest for the 15–19 cohort and declines for older groups, consistent with lower economic and social costs of political participation among younger individuals.

Because the youth share may be endogenous, the baseline relationship cannot be interpreted causally. To address this concern, we implement an instrumental variable strategy based on lagged fertility. The underlying idea is that past fertility realizations—measured using own-country fertility lagged by 15 years—shape the size of youth cohorts many years later, while being predetermined with respect to contemporaneous political changes. Under this assumption, lagged fertility affects democratic outcomes only through its impact on the age structure of the population. A potential concern, however, is that fertility decisions may themselves be influenced by unobserved factors that also shape future political outcomes. To mitigate this issue, we exploit variation in fertility originating from neighboring countries. Specifically, we instrument the youth share using fertility rates in geographically proximate countries, lagged by approximately 20 years, leveraging evidence on the diffusion of fertility behaviors across borders (Daudin et al., 2019; Spolaore and Wacziarg, 2022). This approach isolates demographic variation that is less likely to be directly related to domestic political conditions. Our IV estimates are broadly consistent with the OLS results, although somewhat larger in magnitude. While these findings bring us closer to a causal interpretation, we interpret them with caution, as the validity of the exclusion restriction cannot be fully established.

Finally, we provide complementary evidence based on the timing of demographic transitions. Exploiting the fact that the youth share peaks during specific phases of the transition, we examine whether democratic improvements are more likely to occur around these periods. This approach allows us to incorporate the dynamic dimension

of both demographic and political changes, by studying how the likelihood of democratic improvement evolves before, around, and after the emergence of large youth cohorts. While it does not rely on a sharp treatment definition and should therefore be interpreted as descriptive rather than causal, it provides useful evidence on the timing and dynamics of the relationship between demographic structure and political change, and helps assess the plausibility of the underlying mechanism.

We establish several additional results. First, consistent with the AR theory of democratization, we find that democratic improvements are more likely to occur during recessions. Moreover, the effect of the youth share is amplified in such periods, highlighting a complementarity between economic and demographic forces. Second, we examine the role of life expectancy. While the model predicts an ambiguous effect, our empirical results point to a dominant compositional channel: increases in survival—especially at older ages—are associated with a lower probability of democratic improvement, consistent with a shift in population structure toward older cohorts and a reduction in revolutionary pressure. Third, we provide evidence on the “revolutionary threat” mechanism by studying the relationship between youth cohorts, low-intensity conflicts, and democratic change. We find that a higher youth share is associated with more frequent episodes of unrest, and that these episodes are in turn linked to a higher likelihood of democratic improvement. Finally, we conduct a series of robustness checks and placebo tests that further support our main findings.

Our paper contributes to several strands of the economics and political science literature. First, we contribute to the empirical literature on the determinants of democratization. This literature has primarily focused on economic and social factors, including income inequality (Knutsen and Wegmann, 2016; Haggard and Kaufman, 2012; Dorsch and Maarek, 2020; Kotschy and Sunde, 2021), macroeconomic fluctuations (Acemoglu and Robinson, 2001; Burke and Leigh, 2010; Brückner and Ciccone, 2011), and economic development (Barro, 1999; Lipset, 1959; Murtin and Wacziarg, 2014; Przeworski and Limongi, 1997). By contrast, demographic factors have received relatively limited attention (Dyson, 2013; Wilson and Dyson, 2017). Existing studies typically focus on median age and document a positive association with democratization. We complement this literature by emphasizing the role of the youth share, which captures a distinct and inherently dynamic aspect of demographic change. Unlike median age, the youth share follows a rise-and-fall pattern over the course of the demographic transition, which aligns naturally with the “window of opportunity” mechanism emphasized in Acemoglu and Robinson (2001, 2006). Our analysis, based on a broader country coverage and longer time span, documents a robust relationship between youth cohorts and democratic improvements, and provides suggestive evidence on its underlying mechanisms.

More broadly, our results extend the “window of opportunity” mechanism of Acemoglu and Robinson (2001, 2006) to demographic factors. We show that increases in the youth share can generate conditions similar to those created by economic downturns, by lowering the costs of political mobilization and strengthening the threat of unrest. In this perspective, demographic and economic forces are complementary: the effect of economic shocks on democratization is amplified when the youth share is high. While existing empirical work has focused primarily on economic shocks (Aidt and Franck, 2015; Burke and Leigh, 2010; Brückner and Ciccone, 2011), our results highlight the importance of demographic dynamics in shaping these windows of opportunity.

Second, we relate to the literature on the political consequences of youth bulges. This literature has primarily focused on conflict, social unrest, and political instability (Goldstone, 2002, 2016; Urdal, 2006; Weber, 2013; Flückiger and Ludwig, 2018; Apolte and Gerling, 2018). Consistent with these findings, we show that larger youth cohorts are associated with higher levels of low-intensity unrest. We extend this perspective by showing that such unrest can also be linked to

democratic improvements, suggesting that it may reflect a credible threat that induces elites to concede institutional change.

Finally, our results speak to the modernization hypothesis (Lipset, 1959). While previous work has emphasized the role of education and human capital accumulation as drivers of democratization (Galor and Moav, 2006; Galor et al., 2009; Murtin and Wacziarg, 2014; Parente et al., 2022), we highlight demographic change as a complementary channel. In this perspective, demographic transitions—through their impact on the size and composition of younger cohorts—play a key role in shaping the timing and likelihood of democratic reforms.

The rest of the paper is organized as follows. Section 2 presents the theoretical framework. Section 3 describes the data and provides preliminary evidence. Section 4 outlines the empirical strategy. Section 5 presents the main results, while Section 6 reports additional findings. Section 7 provides robustness checks, and Section 8 concludes.

2. A stylized model

To structure our analysis and guide the empirical investigation, we introduce a simple analytical framework based on the canonical rational choice model of democratization (Acemoglu and Robinson, 2001, 2006). Since the model's primary purpose is illustrative — aimed at clarifying the role of age structure in the democratization process — we adopt a highly stylized version of the AR framework, drawing inspiration from Shadmehr and Haschke (2016) which, to the best of our knowledge, is the unique theoretical paper analyzing the relationship between age structure and threat of revolution.¹

2.1. The model

The demographic structure. Time is discrete indexed by $t \in \{0, \dots, \infty\}$. An overlapping generations economy is composed of two political groups, $g \in \{e, c\}$: the *elite* (e) and the *citizens* (c). Each citizen lives for at most three periods. The first period of life includes childhood and young adulthood. During childhood, individuals are economically inactive; they then become active when entering the young stage (y). We interpret young adulthood as a transitional phase between childhood and full maturity: young individuals are economically and politically active, but not fully integrated into society, with more limited access to formal labor market opportunities and without having formed a family. In the second period, individuals are mature (m), at which point they are fully integrated into the labor market and have children. In the third period, individuals are old (o) and are assumed to be retired. Each mature citizen has $1+n$ children, implying a constant population growth rate n . Mature individuals survive to the third period with probability π , which is assumed to be constant over time. This demographic and life-cycle structure is illustrated in Fig. 1.

Denoting $N_{i,t}$ the number of citizens of age $i \in \{y, m, o\}$ at date t , this demographic structure implies that $N_{y,t} = (1+n)N_{m,t}$, $N_{m,t} = N_{y,t-1}$ and $N_{o,t} = \pi N_{m,t-1}$. Then, denoting $\psi_{i,t} = N_{i,t}/(N_{y,t} + N_{m,t} + N_{o,t})$ the share of citizens of age i in the total citizen population at date t , we get that

$$\psi_{y,t} = \frac{1+n}{2+n+\pi/(1+n)} =: \psi_y, \quad \psi_{m,t} = \frac{1}{2+n+\pi/(1+n)} =: \psi_m \quad (1)$$

and $\psi_{o,t} = 1 - \psi_{y,t} - \psi_{m,t}$. For simplicity, we assume that members of the elite live only one period and that their population size is negligible compared to the citizen population. This simplifying assumption allows us to abstract from the age structure of the elite and the distribution of political power among its different age groups.

¹ In the model, democratization is a discrete transition from autocracy to democracy. In the data, we measure democratization as an increase in a continuous democracy index, capturing gradual institutional reforms. This maps naturally to the model: facing revolutionary threats, incumbents expand political rights and representation. In the model, this is stylized as a full transition; in the data, as incremental democratic concessions.

Political institutions and individual payoffs. The political regime of the economy at time t (denoted P_t) can be one of three types: autocracy A , democracy D , or a revolutionary regime R . The payoff flow received at date t by an individual of group g and age i depends on the current regime P_t and is denoted $w_i^g(P_t)$.

Under autocracy, the elite can use the state's coercive power to appropriate a share of citizens' output, making autocracy the preferred regime for the elite. Moreover, disturbances during a revolutionary period are assumed to be sufficiently severe that the payoff flow for any individual, regardless of group or age, is lower under a revolutionary regime than under any other political state. Therefore, elite payoffs satisfy $w^e(A) > w^e(D) > w^e(R)$.²

To highlight the role of population age structure (ψ_y and ψ_m) in shaping the probability of regime change, we provide a more detailed description of how citizens' payoffs vary over the life cycle. At each date, citizens receive gross income $\tilde{\omega}_i = \omega_i + \varepsilon$.³ The component ω_i is time-invariant and may depend on age $i \in \{y, m, o\}$, capturing persistent cohort-specific differences in economic integration. In contrast, ε captures transitory macroeconomic fluctuations common across cohorts. For tractability, we assume that shocks affect all age groups symmetrically and have zero mean, $\mathbb{E}[\varepsilon] = 0$. We interpret $\tilde{\omega}_i$ as gross income prior to political extraction or destruction; political institutions determine how much of this income is ultimately retained by individuals.

Under autocracy, a fraction $\theta \in (0, 1)$ of each citizen's gross income is expropriated by the ruling elite. Under democracy, no group can coercively appropriate others' output, so each citizen retains their full gross income. During a revolutionary period, a fraction $\mu \in (0, 1)$ of citizens' gross income is destroyed. A revolution also entails an age-specific fixed disutility cost $\chi_i > 0$, capturing psychological and non-economic costs associated with political unrest. Citizens' payoffs are therefore given by

$$w_i^e(A) = (1 - \theta)\tilde{\omega}_i, \quad w_i^c(D) = \tilde{\omega}_i, \quad w_i^c(R) = (1 - \mu)\tilde{\omega}_i - \chi_i. \quad (2)$$

As discussed above, we assume that citizens receive a higher payoff flow under autocracy than during a revolution, which implies $\mu > \theta$. We also impose the following assumptions on age-specific earnings and revolutionary disutility.⁴

Assumption 1. $\omega_y < \omega_m$ and $\chi_y < \chi_m$.

The assumption $\omega_y < \omega_m$ reflects that young adulthood is a transitional phase between schooling and full labor market integration. In many countries, young individuals face significant barriers to entry into stable and formal employment, while older cohorts occupy more protected positions (Lindbeck and Snower, 2001; Saint-Paul, 2002). Consistent with this view, youth unemployment rates are typically higher than aggregate unemployment, and young workers disproportionately hold low-quality and informal jobs (Lee, 2013).

The assumption $\chi_y < \chi_m$ captures age differences in the perceived costs of political unrest. In the model, only mature individuals have children, so younger individuals do not yet bear family responsibilities and may therefore perceive revolutionary disruptions as less costly. Moreover, evidence from developmental psychology indicates that the transition from childhood to adulthood is characterized by lower sensitivity to negative outcomes and altered risk perception (Casey et al.,

² These payoffs can be microfounded through assumptions on initial endowments, production technologies, and fiscal institutions, as in Acemoglu and Robinson (2006).

³ For expositional simplicity, we index productivity by age group $i \in \{y, m, o\}$, although the first life period comprises childhood and young adulthood (see Fig. 1). Hence, ω_y should be interpreted as productivity during the active sub-period of the first life stage.

⁴ Note that we do not need to impose any restriction on ω_o and χ_o for the main results.

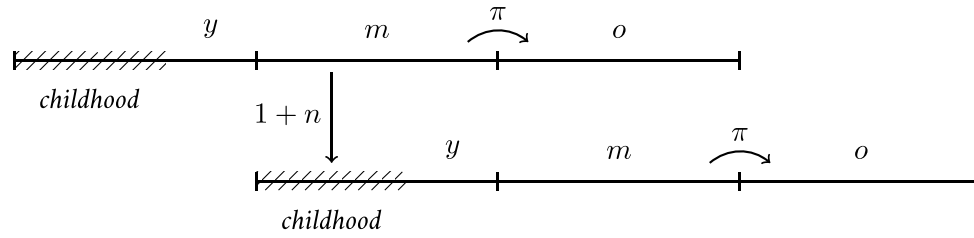


Fig. 1. Demographic and life-cycle structure of the citizen population.

2008; Steinberg, 2017). Relatedly, Fitzenberger et al. (2022) document that younger cohorts exhibit lower risk aversion and greater openness to new experiences than older cohorts.

Solving the model. Since all members of the elite, as well as all citizens of a particular age group, share identical preferences, we treat these groups collectively. Thus, we refer to *the elite* as one group, and to *the young*, *the mature*, and *the old* as distinct age-based groups among citizens.⁵ We then examine an equilibrium that results from the interactions between these groups.

Since our focus is on the democratization process, we consider a society initially governed by an autocracy. We assume that if a democracy is established at any given time t , the society will remain democratic for all subsequent dates $t' \geq t$. If the political regime is non-democratic at time t , the elite may choose to initiate a transition to democracy.⁶ If the elite does not, some citizens may decide to participate in a democratic movement. Let ρ_t represent the fraction of citizens who find it beneficial to join a democratic movement. We assume that this movement leads to a revolution if and only if $\rho_t \geq \zeta$, where ζ is the realization of a random variable with cumulative distribution function denoted $\phi(\cdot)$. The random variable ζ captures uncertainty in the success of the revolutionary movement, reflecting the role of unforeseen circumstances in determining whether a given level of participation leads to a revolution. We assume that the success of a revolution depends on the proportion of democratic activists within the population rather than their absolute number (see, e.g. Persson and Tabellini, 2009; Shadmehr, 2019; Bueno de Mesquita and Shadmehr, 2022). Revolution is a temporary state lasting one period, after which the economy reaches D , an absorbing state.

Assuming that, at the beginning of a period t , the political regime is autocratic ($P_t = A$), the events of the period t unfold as follows:

1. nature chooses ε and then ζ ;
2. the elite decides to establish a democracy or not;
3. if the elite decided not to establish a democracy in stage 2, each group decides to take part in a democratic movement or not.

Individuals make their decisions in order to maximize their expected flow of income over their entire life. For simplicity, we assume that individuals' expectation with respect to the political regime that will prevail in the future are myopic.⁷ If at date t the elite does not democratize, individuals anticipate that the political regime that will prevail in $t' > t$ will be non-democratic if a revolution does not occur during the period t . The problem is solved by backward induction. We first look at the decision to participate or not in a democratic movement when the elite has decided not to democratize (Stage 3), then we analyze the decision of the elite to democratize or not (Stage 2).

Participation in a democratic movement (Stage 3). Citizens of age group i participate in a democratic movement if and only if their life-cycle utility under a successful revolution exceeds their utility in the absence of revolution. This participation condition, which we refer to as the *revolution constraint*, is given by

$$(1-\theta)(\tilde{\omega}_i + W_i) \leq (1-\mu)\tilde{\omega}_i + W_i - \chi_i \Leftrightarrow \theta W_i \geq (\mu-\theta)(\omega_i + \varepsilon) + \chi_i, \text{ (RC}_i\text{)}$$

where W_i denotes the expected present value of future income for a citizen of age i . Given the life-cycle structure, $W_o = 0$, $W_m = \mathbb{E}[\pi\tilde{\omega}_o] = \pi\omega_o$, and $W_y = \mathbb{E}[\tilde{\omega}_m + \pi\tilde{\omega}_o] = \omega_m + \pi\omega_o$.

Condition (RC _{i}) has a straightforward interpretation. Participation in a democratic movement trades off the future gains from a transition to democracy, θW_i , against the net costs of revolutionary activity, $(\mu-\theta)(\omega_i + \varepsilon) + \chi_i$.⁸ Since $\mu > \theta$ and $\chi_i > 0$, these costs are strictly positive. Eq. (RC _{i}) clarifies how age shapes participation incentives and, in turn, how demographic structure affects the probability of regime change. Age affects the benefits of participation through a horizon effect: younger individuals enjoy a longer expected duration of democracy, implying $W_y = \omega_m + \pi\omega_o > W_m = \pi\omega_o > W_o = 0$. Age also affects participation costs through contemporaneous opportunity costs ω_i and the disutility of revolution χ_i . Under Assumption 1, both channels make the revolution constraint less binding for younger cohorts. It follows immediately that the revolution constraint is never satisfied for the old, i.e., (RC _{o}) never holds. Hence, old individuals never find it optimal to participate in a democratic movement: the horizon effect is insufficient to offset the contemporaneous costs of revolutionary activity.

For mature and young individuals, condition (RC _{i}) is satisfied whenever μ is below an age-specific threshold $\bar{\mu}_i$, given by

$$\bar{\mu}_y = \theta + \frac{\theta(\omega_m + \pi\omega_o) - \chi_y}{\omega_y + \varepsilon} \quad \text{and} \quad \bar{\mu}_m = \theta + \frac{\theta\pi\omega_o - \chi_m}{\omega_m + \varepsilon}.$$

⁵ Since revolutionary activities generate collective benefits while imposing private costs, they typically give rise to a collective action problem. Following AR, we abstract from this issue and treat both the elite and groups of citizens as unitary actors. Acemoglu and Robinson (2006) provide an extensive discussion of how revolutionary leaders may overcome collective action problems, including by offering private benefits to participants (Tullock, 1971). This mechanism could be explicitly introduced by assuming that non-participants are, to some extent, excluded from the benefits of regime change.

⁶ To focus on the role of age structure in the transition to democracy, we set aside other strategies that the elite might employ when facing the threat of revolution. In particular, we do not account for the possibilities of repression or economic concessions (e.g., income redistribution). These strategies are explicitly analyzed in Shadmehr and Haschke (2016). At the heart of the AR's window of opportunity theory of democratization is the idea that such alternative strategies may be ineffective. Specifically, economic concessions face a commitment problem: the elite may promise redistribution to appease the revolutionary threat but, once the immediate danger is over, they have incentives to renege on these promises. Without credible enforcement mechanisms, the citizens have little reason to trust these concessions, making democratization a more viable path for securing long-term political and economic rights. This is especially true when the threat of revolution is temporary, as in the case of an economic recession or, to some extent, a transitory increase in the youth share during the demographic transition.

⁷ In Online Appendix O.1, we relax this assumption and, within a slightly simpler framework, demonstrate that our results can be derived as the Subgame-Perfect Nash Equilibrium of a game in which citizens are forward-looking.

⁸ Under myopic expectations, individuals assume that, absent a revolution today, the political regime remains autocratic indefinitely.

Under [Assumption 1](#), we have $\bar{\mu}_y > \bar{\mu}_m$, implying that young individuals are more prone to participate in revolutionary activity. As discussed above, this difference is likely to be quantitatively large, since young adulthood is characterized by both a lower opportunity cost of participation (ω_y) and a lower disutility cost of revolution (χ_i). Accordingly, young adults are substantially more likely to engage in revolutionary actions, consistent with existing empirical evidence ([Nordås and Davenport, 2013](#); [Weber, 2013, 2021](#)).

To sharpen this distinction, we impose the following assumption:

Assumption 2. $\chi_m > \theta\pi\omega_o$.

Under [Assumption 2](#), mature individuals never participate in revolutionary activity. Consequently, the fraction ρ_t of citizens who choose to participate in a revolution equals 0 if $\mu > \bar{\mu}_y$ and ψ_y if $\mu \leq \bar{\mu}_y$.

We can thus express the probability of a successful revolution—conditional on the realization of ε but prior to the realization of ζ —as a function of the youth share ψ_y :

$$P(\psi_y) = \begin{cases} \phi(\psi_y) & \text{if } \mu \leq \bar{\mu}_y, \\ 0 & \text{if } \mu > \bar{\mu}_y, \end{cases} \quad (3)$$

where $\phi(\cdot)$ is increasing in ψ_y .

Decision of the elite (Stage 2). Since the elite know the realization of ε and ζ when they have to choose to democratize or not they can perfectly anticipate if a successful revolution will occur or not. If it is not the case, they do not have any incentive to democratize while, if it is the case, they always have incentives to do so (indeed, $w^e(\mathcal{A}) > w^e(\mathcal{D}) > w^e(\mathcal{R})$). Hence, the probability of a (voluntary) transition from non-democracy to democracy is exactly given by $P(\psi_y)$ and a revolution never occurs in equilibrium.

The result that the autocratic elite always concedes democracy peacefully before any civil disorder arises is largely driven by the assumption of perfect information on both the elite's and the citizens' sides. In [Appendix A](#), we extend the model by relaxing this assumption, allowing for a scenario in which citizens are better informed than the elite about the costs of participating to revolutionary activities. Additionally, we introduce the idea that low-intensity civil conflicts, such as protests, serve as a necessary precursor to a full-scale revolution. In this extended model, when the youth share is sufficiently high and the costs of protests and revolution are low, it becomes optimal for young individuals to participate in protests. These protests, in turn, act as signals that reveal information about the state of the world to the elite, creating incentives for them to concede democracy.

2.2. Discussion and testable predictions

Eq. (3) characterizes the demographic determinants of democratic transition. The threat of revolution is higher in societies with a larger youth share, reflecting three complementary mechanisms. First, a horizon effect implies that younger cohorts, with a longer expected political horizon, place greater value on future democratic benefits. Second, lower contemporaneous opportunity costs of political mobilization among younger individuals, due to weaker labor market integration, reduce the cost of participation. Third, lower perceived disutility costs of revolutionary activity further increase their willingness to mobilize. Through this combination of channels, a temporary increase in the youth share—such as during a demographic transition—can generate heightened revolutionary pressure and induce political concessions toward democracy. While the identity of the most weakly integrated sub-cohort within the youth population may vary across contexts, it is necessarily concentrated among the youngest cohorts. Given the coarse structure of the available demographic data, which is organized in five-year age groups, the empirical counterpart of this “young” category is most closely captured by the 15–19 age group, while older cohorts correspond to progressively more economically and socially integrated individuals.

This reasoning yields the following testable prediction of the model.

Prediction 1. A rise in the youth share increases the probability of a democratic improvement.

Demographic factors are complemented by economic factors which are at the heart of the AR theory of democratization. In particular, it is clear from (RC₁) that a temporary fall in economic opportunities (captured by a lower ε) will incentivize more people to participate in a democratic movement by decreasing the opportunity costs of participation. This effect might have a large impact on the probability of democratization. This is the case if, for instance, such a decrease pushes $\bar{\mu}_y$ above μ such that the probability of a revolution switches from 0 to $\phi(\psi_y)$. Moreover, the impact of this temporary decrease of opportunity costs is magnified when the youth share is large (since $\phi(\psi_y)$ is increasing in ψ_y). This is our second main prediction.

Prediction 2. The positive impact of depressed economic opportunities on the probability of a democratic improvement is amplified when the youth share is large.

Other factors that shape the relative costs and benefits of revolution also affect the likelihood of democratic improvement. In particular, life expectancy, proxied by the survival probability π in the model, strengthens the horizon effect. An increase in life expectancy raises the expected duration over which young individuals benefit from a regime change, thereby increasing the gains from democratization. Formally, a higher π relaxes the revolution constraint for the young, such that $\bar{\mu}_y$ is increasing in π . However, longer life expectancy also affects the probability of democratic improvement through its impact on demographic structure. As life expectancy rises, the youth share declines (formally, ψ_y is decreasing in π in Eq. (1)), which mechanically reduces revolutionary pressure. Consequently, the overall effect of life expectancy on democratic improvement is theoretically ambiguous and must be assessed empirically.

Another crucial factor is the time-invariant component of youth income, ω_y , which captures the opportunity cost of political mobilization and therefore negatively affects $\bar{\mu}_y$. Conversely, the time-invariant component of mature individuals' income, ω_m , increases the expected gains from democratization for young individuals and thus positively affects $\bar{\mu}_y$. Hence, the lower is ω_y relative to ω_m , the higher is the probability of democratic improvement and the stronger is the effect of the youth share on this probability. This complementarity channel is of particular empirical interest because the ratio ω_y/ω_m can be interpreted as a measure of the relative economic integration of younger cohorts, which lies at the core of our theory linking youth bulges to democratic change. Unfortunately, we lack consistent long-run cross-country proxies for this ratio. For example, youth unemployment data are available for a large set of countries only from the early 1990s, which limits the feasibility of a systematic empirical test of this mechanism. Existing studies nevertheless document an interaction between the youth share and labor market conditions in explaining civil conflict and protest activity ([Campante and Chor, 2012](#); [Apolte and Gerling, 2018](#); [Gerling, 2018](#)). Those findings are consistent with the model's prediction that lower relative economic opportunities for youth amplify the political consequences of a large youth cohort. Finally, although ω_y and ω_m are treated as exogenous in the model, relative youth economic opportunities may in practice decline with the youth share through standard supply-and-demand mechanisms.⁹ To the extent that this is the case, part of the effect of ω_y/ω_m is already embedded in the estimated impact of the youth share. This additional channel would therefore reinforce our first prediction of a positive effect of the youth share on the probability of democratic improvement.

⁹ A large literature documents cohort-size effects on youth labor market outcomes (“cohort crowding”), whereby larger youth cohorts experience higher unemployment or lower relative earnings ([Bloom and Freeman, 1986](#); [Zimmermann, 1991](#); [Korenman and Neumark, 1997](#); [Sun et al., 2023](#)).

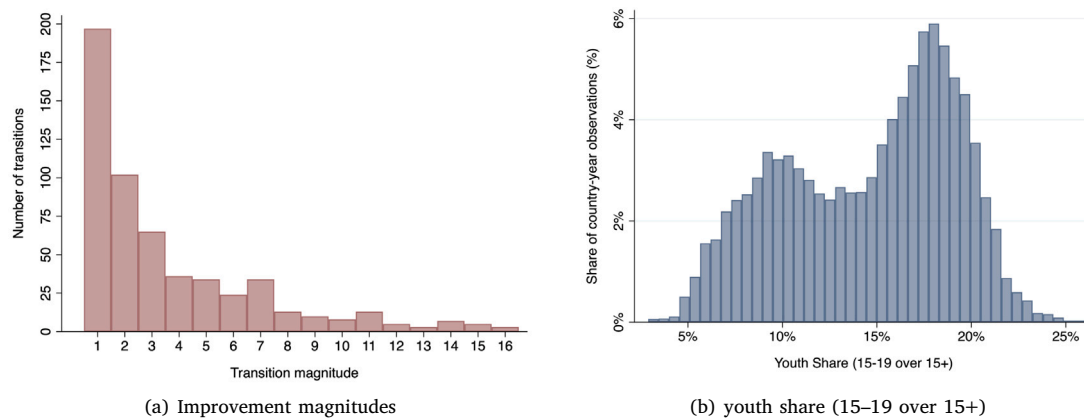


Fig. 2. Distribution of key variables.

Notes: Panel (a) reports the distribution of positive year-to-year changes in the Polity2 score, measured as the number of transition episodes for each magnitude. Panel (b) reports the distribution of the youth share, defined as the share of individuals aged 15–19 in the population aged 15 and above, expressed as the share of country-year observations (in percent). Both panels are based on country-year observations in the baseline sample (1950–2018).

Beyond labor market conditions and other determinants of ω_y , the opportunity cost of political mobilization may also be affected by the possibility for the elite to implement targeted policies toward specific groups, in particular younger cohorts. For instance, policies aimed at improving youth economic conditions could raise ω_y and thereby reduce incentives to participate in political unrest. In our framework, we abstract from such responses and treat ω_y and ω_m as exogenous. This simplification is in line with Acemoglu and Robinson (2006), where redistributive concessions are subject to a commitment problem and may therefore be insufficient to credibly offset revolutionary threats when these are large. At the same time, Shadmehr and Haschke (2016) highlight that the ability of the state to target concessions or repression toward specific groups — in particular younger cohorts — can substantially shape political outcomes. This perspective is closely related to the mechanism we emphasize.

Finally, the result that the elite always peacefully concedes power before the onset of any civil disorder hinges on specific modeling assumptions. In particular, in the absence of uncertainty, the threat of revolution is perfectly foreseeable to the elite, who can precisely infer whether a revolution will occur. As illustrated in the extensions presented in Appendix A, when uncertainty is introduced, civil disorder, protests, or riots may instead play a role in the democratization process by making the revolutionary threat more credible in the eyes of the elite (Andrews and Jackman, 2005; Aidt and Leon, 2016). This perspective helps reconcile our results with the political science literature emphasizing the role of young people in riots and civil violence. Taken together, these considerations lead us to formulate the prediction that low-intensity conflicts, such as riots, may act as a transmission channel through which a high youth share increases the probability of a democratic improvement.

3. Data and preliminary evidence

In this section, we describe the data we employ and provide some descriptive raw evidence of the relationship between demography and democratization.

3.1. Data

This section presents the data used in the empirical analysis and introduces our key variables. We begin by describing our main outcome variable capturing democratic improvements. We then present the demographic variables at the core of our analysis and finally describe the additional control variables.

Democratic improvements. Our main outcome variable captures episodes of democratic improvement using the Polity2 score from Marshall et al. (2019). The Polity2 index measures the degree of democracy on a scale ranging from -10 (full autocracy) to $+10$ (full democracy). It summarizes several institutional characteristics of political regimes, including the competitiveness and openness of executive recruitment, constraints on the executive, and the regulation and competitiveness of political participation (see Online Appendix O.7 for more details).

Fig. 2a presents the distribution of positive year-to-year changes in the Polity2 score. Two features of the data are particularly noteworthy. First, democratic improvements frequently occur through relatively small changes in the index: a substantial share of transitions correspond to increases of only one point. This pattern suggests that institutional reforms often involve gradual adjustments, rather than exclusively occurring through large discrete jumps. Second, the Polity2 index is bounded above at $+10$, which mechanically limits the scope for institutional improvement among country-years located near the top of the distribution. Consistent with this constraint, Appendix Fig. 11 shows that the probability of a democratic improvement (measured by an increase in the Polity2 score) declines sharply as the Polity2 score approaches its upper bound.

Motivated by these considerations, we define our baseline democratic improvement indicator as a binary variable equal to one when the Polity2 score increases by at least one point between year $t - 1$ and year t , and zero otherwise. This definition allows us to capture incremental institutional changes that would be missed by more restrictive thresholds.¹⁰ It also helps mitigate distortions induced by the bounded nature of the Polity scale, which compresses the distribution of potential improvements near its upper bound. In addition, given the relatively low frequency of democratic improvements in long-run country-year panels, this definition yields a less sparse and more informative outcome variable.

To ensure that the changes we capture correspond to sustained institutional improvements rather than short-lived fluctuations in the index, we refine this indicator by imposing a persistence condition: the Polity2 score must remain strictly higher than its level in year $t - 1$ during the three years following the transition. Our baseline outcome variable therefore captures episodes of *consolidated democratic improvement*. In Section 7.2, we show that our results are robust to alternative

¹⁰ The Online Appendix O.7 provides illustrative examples of one-point increases in the Polity2 score, illustrating the political events captured by this baseline definition.

measures of democratic improvements, including those based on other democracy indices such as the *polyarchy* index from the V-Dem dataset (Coppedge et al., 2021) and the Machine Learning Democracy Index (Gründler and Krieger, 2016, 2021).¹¹

Demographic variables. Our main explanatory variable captures the relative size of youth cohorts in the population. Using age-specific population data from the United Nations World Population Prospects (United Nations, Department of Economic and Social Affairs, Population Division (2022), 2022), we define the *youth share* as the share of individuals aged 15–19 in the population aged 15 and above. This definition is guided by the theoretical distinction between “young” and “mature” individuals developed in Section 2, where the youngest cohorts are those least integrated into the labor market and family structures.

Fig. 2(b) shows that the distribution of this variable is bi-modal, with one mode at relatively low youth shares and another at substantially higher values, consistent with countries being at different stages of the demographic transition. As shown in Appendix Fig. 12, the distribution shifts over time: the high youth share mode gradually shrinks while the lower youth share mode becomes increasingly prominent. This pattern reflects the well-known dynamics of demographic transitions, during which countries experience temporary “youth bulges” before the relative size of younger cohorts declines.

We complement this measure with additional demographic indicators derived from fertility and age-specific mortality data from the same United Nations World Population Prospects source. These data provide measures of net fertility, which we use to construct an instrument for the youth share. Age-specific mortality data allow us to compute survival probabilities that we exploit in several specifications.

We also rely on the dataset constructed by Delventhal et al. (2021), which estimates the timing of demographic transitions across countries using multiple historical sources. Following the standard model of demographic transition, we interpret the peak of the transition — corresponding to the maximum of net fertility — as the point at which fertility begins to decline. Because cohorts born at the fertility peak enter the 15–19 age group approximately 17 years later, we shift this date forward by 17 years to approximate the period during which youth cohorts are largest relative to the adult population. We then construct a binary indicator identifying a window centered around this shifted transition date, which we exploit in panel specifications with country and year fixed effects. This approach provides a complementary way to capture the demographic conditions associated with large youth cohorts that underlie our main empirical specification based on the youth share.

Control variables. Our baseline specifications include a set of standard controls capturing structural determinants of political institutions. Data on GDP per capita come from Maddison Project Database (2020). Urbanization rates are obtained from the United Nations (2022). Additional country-level control variables are drawn from the Quality of Government dataset (Teorell et al., 2020), while data on manufacturing employment shares are taken from van Noort (2024). For further analyses and robustness checks, we also use data on political unrest — including riots and revolutions — from the Cross-National Time-Series data archive (Banks, 2020).

Summary statistics. Table 1 reports summary statistics for the main variables used in the baseline analysis, including our measure of democratic improvement, the youth population ratio, and standard economic and institutional controls.

¹¹ In some cases, the Polity2 score relies on coding rules that may imperfectly capture political transitions. In particular, a value of zero is assigned during interregnum periods, which may generate spurious improvements when standard coding resumes. Following Brückner and Ciccone (2011) and Aidt and Leon (2016), we adopt a conservative approach and exclude these observations. We also show that our results are robust to alternative coding rules (see Online Appendix O.2).

3.2. Preliminary evidence

To provide a first illustration of the relationship between demographic structure and democratic change, we begin by examining simple graphical patterns in the data.

Fig. 3 presents a binned representation of the relationship between the youth share and the probability of democratic improvement.¹² Panel (a) shows a strong positive relationship in the pooled country-year data: observations with larger youth cohorts are associated with a higher probability of democratic improvement. Panel (b) focuses on within-country variation over time by removing country-specific average levels from both variables, so that each observation corresponds to deviations from a country’s own mean. The positive relationship persists when considering only within-country fluctuations. Although the youth share exhibits substantially less variation within countries than across countries (see the within and overall standard deviations in Table 1), periods in which it is relatively high compared to a country’s own average are associated with a higher probability of democratic improvement.

We next examine how demographic transitions and political transitions evolve over time across groups of countries. Fig. 4 shows the evolution of the youth share and the probability of democratic improvement for groups of countries defined by their initial level of GDP per capita. Countries in higher income quartiles exhibit systematically lower youth shares, consistent with their more advanced stage in the demographic transition. Across groups, the youth share follows a hump-shaped pattern, increasing before gradually declining. For the highest income quartile, this decline is already well underway at the beginning of the sample period, indicating that these countries have largely completed their demographic transition.

For the three lowest quartiles of development, increases in the youth share tend to coincide—often with a short lag—with increases in the probability of democratic improvement, while the subsequent decline in the youth share is accompanied by a reduction in the frequency of democratic improvements. This temporal co-movement suggests that periods characterized by large youth cohorts may be associated with increased political pressure and institutional change. Although these aggregated trends conceal heterogeneity in the timing of demographic transitions across individual countries, they point to a systematic association between youth bulges and episodes of democratic improvement.

As a complementary way to visualize the role of demographic transitions, Appendix Fig. 14 displays, for each country, the timing of the demographic transition window together with the occurrence of democratic improvements. The figure shows that democratic improvements occur more frequently during periods associated with large youth cohorts. Consistent with this visual pattern, the average probability of a democratic improvement is 0.077 within the transition windows, compared to 0.043 outside the windows. This simple comparison suggests that the demographic conditions captured by the transition window are associated with a substantially higher likelihood of democratic improvement and provides a direct visual counterpart to our empirical strategy based on the timing of demographic transitions. These graphical patterns provide suggestive evidence of a positive relationship between youth bulges and democratic improvement, both across countries and within countries over time. While these patterns are purely descriptive and do not establish causality, they provide consistent motivation for the empirical analysis that follows. In the next section, we investigate this relationship more systematically using econometric specifications that exploit within-country variation in demographic structure.

¹² A raw scatterplot of the data is reported in Appendix Fig. 13.

Table 1
Summary statistics (Baseline sample, 1950–2018).

	Obs.	Mean	Std.Dev.	Std.Dev. (within)	Min.	Max.
Democratic improvement	9210	0.050	0.218	0.214	0	1
Youth share (15–19)	9107	0.151	0.045	0.021	0.04	0.26
ln(GDP p.c.)	8688	8.627	1.156	0.514	6.14	11.96
Growth rate (GDP p.c.)	8663	2.449	5.889	5.706	–100	105.44
Polity2	9210	0.920	7.510	4.560	–10	10

Notes: The unit of observation is a country-year. The sample includes observations between 1950 and 2018 for which the dependent variable, *consolidated democratic improvement*, is observed. Democratic improvement is a binary indicator equal to one if a country experiences a consolidated improvement in its level of democracy in a given year. The youth share measures the share of the population aged 15–19 relative to the population aged 15 and above. GDP per capita is expressed in natural logarithms, and growth corresponds to the annual percentage change in GDP per capita. Polity2 is the democracy score from the Polity IV dataset. The column “Std. Dev. (within)” reports the within-country standard deviation.

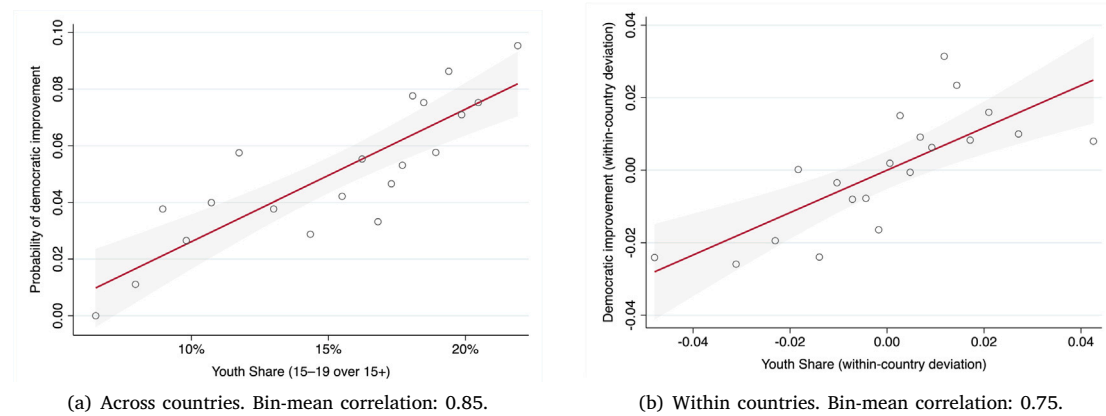


Fig. 3. Youth share and democratic improvement.

Notes: Each dot represents the average of observations within 20 equally sized groups of similar values of the youth share. The solid line shows the fitted linear relationship. In panel (b), both variables are measured relative to their country-specific average so that the relationship is identified using only changes within countries over time.

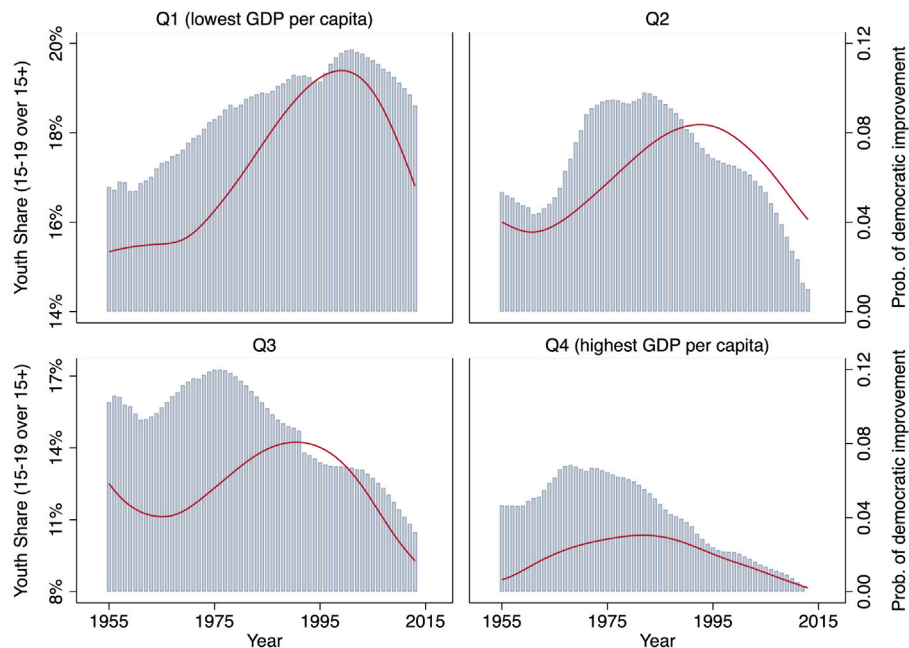


Fig. 4. Youth share and democratic improvement over time by income group.

Notes: Countries are grouped into quartiles based on their GDP per capita in 1990 and remain in the same group throughout the sample period. Bars represent the average youth share (share of individuals aged 15–19 relative to the population aged 15 and above) in each group and year. The red line shows a locally weighted regression (LOWESS) of the probability of democratic improvement on time, which smooths short-run fluctuations to highlight medium-run trends. The LOWESS curves are estimated using the full sample (1950–2018) but displayed over the 1955–2013 period to mitigate boundary effects. Vertical scales differ across panels.

4. Empirical strategy

This section presents our empirical strategy to study the relationship between demographic structure and democratic improvement. We begin with a baseline panel specification and then discuss extensions that address potential mechanisms and identification concerns.

Baseline specification. We estimate the following panel regression using a Linear Probability Model (LPM):

$$democ_improv_{i,t} = \alpha + \beta \times youth_share_{i,t-1} + X'_{i,t-1} \Gamma + \delta_i + \epsilon_t + u_{i,t}. \quad (4)$$

The dependent variable $democ_improv_{i,t}$ is a binary indicator equal to one when country i experiences a consolidated democratic improvement in year t , as defined in Section 3.

The main independent variable, $youth_share_{i,t-1}$, captures the relative size of youth cohorts. According to our theoretical framework, higher values of this ratio are associated with a higher likelihood of democratic improvement. Country fixed effects δ_i control for time-invariant characteristics that may affect the propensity to democratize, while year fixed effects ϵ_t capture global shocks and common trends, such as waves of democratization. The vector $X_{i,t-1}$ includes time-varying controls that account for potential confounding factors. Throughout, standard errors are clustered at the country level.

Control variables. Our baseline set of controls includes variables commonly identified as determinants of democratization. In particular, we control for GDP per capita, which captures the level of economic development (Lipset, 1959; Przeworski and Limongi, 1997; Acemoglu et al., 2008), and for macroeconomic fluctuations (growth rate of GDP per capita), as adverse economic shocks may facilitate political change (Acemoglu and Robinson, 2001; Burke and Leigh, 2010; Brückner and Ciccone, 2011). We also include the lagged Polity score to account for the fact that the scope for democratic improvement depends on the initial level of political institutions (Burke and Leigh, 2010).

An additional issue arises from the bounded nature of the Polity index. By construction, countries with a Polity score of 10 cannot experience further democratic improvement. Rather than excluding these observations from the sample, which would mechanically alter the composition of the panel, we include a dummy variable equal to one when the lagged Polity score is equal to 10. This control accounts for the fact that the probability of observing a democratic improvement is mechanically zero at the upper bound of the index.

In additional specifications, we consider a broader set of controls, including population size, inequality, education, urbanization, and the structure of production, which have been linked to political development in the existing literature.

Endogeneity and instrumental variables. The youth share may be endogenous to political and economic conditions, as demographic outcomes and institutional change may be jointly determined by underlying development processes. As a result, the estimates from Eq. (4) should be interpreted as conditional correlations rather than causal effects.

To address these concerns, we implement instrumental variable strategies that exploit the delayed mechanical relationship between fertility behavior and the size of youth cohorts. Changes in fertility translate into variation in the size of cohorts reaching adolescence roughly 15 to 20 years later, providing a natural source of variation in the youth share.

Our identification strategy relies on two instruments constructed from lagged fertility dynamics. The first instrument is based on domestic fertility. Specifically, we use the net fertility rate averaged over a five-year window and lagged by 15 years. This smoothing reflects the fact that the youth share is measured using five-year age cohorts (15–19), so that the relevant cohort size depends on fertility realized over several consecutive years. The 15-year lag aligns this fertility measure with the birth years of individuals aged 15–19. This instrument is therefore mechanically correlated with the youth share, while—given

the long lag—being less likely affected by contemporaneous political and economic conditions.

The second instrument exploits fertility patterns in neighboring countries. We construct the following instrument for the youth share in country i of region r at time t :

$$Z^r_{i,t} = \frac{1}{N^r_{i,t} - 1} \sum_{j \in r, j \neq i} \overline{fertility_rate}_{j,t-20}, \quad (5)$$

where $N^r_{i,t}$ denotes the number of countries in region r , and $\overline{fertility_rate}_{j,t-20}$ denotes the net fertility rate in country j , averaged over a five-year window and lagged by 20 years.¹³ This approach captures the gradual diffusion of fertility behaviors across countries and provides variation in domestic demographic structure that is less directly tied to country-specific political conditions. It is similar in spirit to the regional instruments used by Acemoglu et al. (2019) and Dorsch and Maarek (2019), and is motivated by evidence on the geographical diffusion of fertility behaviors (Daudin et al., 2019; Spolaore and Wacziarg, 2022).

These two instruments capture different aspects of the same demographic process. Domestic fertility provides a strong predictor of youth cohort size but may be more directly affected by country-specific factors. In contrast, neighboring fertility relies on external variation and longer lags, which may improve the plausibility of the exclusion restriction, albeit at the cost of a weaker first-stage relationship.

We estimate the corresponding two-stage least squares model:

$$youth_share_{i,t} = \alpha + \beta Z_{i,t-1} + X'_{i,t-1} \Gamma + \delta_i + \epsilon_t + u_{i,t}, \quad (6)$$

$$democ_improv_{i,t} = \alpha + \beta^{2S} \widehat{youth_share}_{i,t-1} + X'_{i,t-1} \Gamma + \delta_i + \epsilon_t + u_{i,t}. \quad (7)$$

All specifications include our baseline set of controls as well as country and year fixed effects.

The identifying assumption is that lagged fertility affects democratic improvement only through its impact on demographic structure. While this assumption cannot be fully tested, we take several steps to assess its plausibility. In particular, the inclusion of a rich set of controls capturing economic conditions and political institutions helps mitigate concerns that the instruments proxy for broader development dynamics. We also conduct a range of additional checks to examine the robustness of the first-stage relationship and to probe potential violations of the exclusion restriction.

Despite these efforts, the validity of the exclusion restriction remains an important limitation in this setting. We therefore interpret the IV estimates with caution and view them as providing complementary evidence rather than definitive causal identification.

Alternative empirical strategies. In addition to the baseline specification, we exploit the timing of demographic transitions using the data described in Section 3. Specifically, we construct an indicator for periods in which countries are expected to experience large youth cohorts, based on the timing of fertility peaks (i.e., the year of maximum net fertility, shifted forward to account for cohort aging, as described in Section 3). This approach provides a complementary way to capture demographic conditions associated with youth bulges without relying directly on detailed age-structure data.

Building on this timing information, we compare the probability of democratic improvement across broad periods before, around, and after the emergence of large youth cohorts. Concretely, we define time intervals relative to the date at which cohorts born at peak fertility reach youth ages, and estimate how the likelihood of democratic improvement varies across these intervals within countries. This approach

¹³ We consider the following regions: Western Europe, Northern Europe, Southern Europe, Eastern Europe, Northern Africa, Western Africa, Middle Africa, Eastern Africa, Southern Africa, Western Asia, Central Asia, East Asia, South-East Asia, South Asia, Oceania (including Australia and the Pacific), North America, Central America, South America and Caribbean.

allows us to assess whether democratic improvements are disproportionately concentrated around demographic transitions, rather than focusing on precise year-by-year timing.

An important advantage of this strategy is that it can be implemented over a longer time horizon. Because it relies on estimated dates of fertility transitions rather than detailed demographic data, it allows us to extend the analysis to earlier periods for which age-structure data are not available. This provides additional variation and statistical power, and helps capture long-run patterns in the relationship between demographic change and political development.

More broadly, this approach provides a complementary non-IV perspective on the relationship between demographic structure and democratization. By organizing the data around the timing of demographic transitions, it helps address concerns that the baseline results may reflect residual persistence in democratic outcomes rather than a systematic association with demographic conditions. While this strategy does not rely on a sharp treatment definition and should therefore be interpreted as descriptive rather than causal, it offers useful evidence on the timing and persistence of the relationship between youth bulges and democratic improvement.

Mechanisms. Finally, we explore mechanisms suggested by our theoretical framework. First, we examine whether the effect of demographic structure depends on macroeconomic conditions by interacting the youth share with measures of economic fluctuations. These specifications allow us to test the prediction that adverse economic shocks have stronger political effects in the presence of large youth cohorts.

Second, we investigate the role of intertemporal incentives by considering survival probabilities as a proxy for the expected horizon over which individuals benefit from institutional change. This allows us to assess whether the relationship between demographic structure and democratic improvement depends on the expected duration of political gains.

Third, we examine the “revolutionary threat” mechanism by studying the effect of the youth share on low-intensity conflicts such as riots. We assess whether increases in the youth population raise the incidence of unrest and whether such unrest is associated with subsequent democratic improvements.

5. Panel regression analysis

5.1. Linear probability model results

Baseline estimates. Table 2 reports estimates of Eq. (4). Our main regressor is the youth share, defined as the share of individuals aged 15–19 in the population aged 15 and above. The dependent variable equals one if the Polity2 score increases by at least one point between $t - 1$ and t and remains above its pre-transition level for at least three subsequent years. All specifications include country and year fixed effects, as well as a ceiling control (a dummy equal to 1 when the Polity2 score in $t - 1$ equals 10). Standard errors are clustered at the country level.

Columns 1–5 sequentially add standard controls used in the literature on democratic improvements. Across these specifications, the estimated coefficient on the youth share is positive and statistically significant. Balancing concerns about omitted-variable bias and sample size, we take column 5 as our baseline specification. In this column, the estimated coefficient is 0.49 (s.e. 0.20), indicating that a larger share of individuals aged 15–19 is associated with a higher probability of democratic improvement.

We next assess robustness to a broader set of socioeconomic controls. Column 6 adds urbanization, log population, average years of schooling among individuals aged 15 and above, the Gini coefficient, and the manufacturing employment share. Including these variables substantially reduces the sample size and increases standard errors. While the coefficient on the youth share remains positive and larger

in magnitude, it is no longer statistically significant. For brevity, we do not report the coefficients on the additional controls in Table 2; they are presented and discussed in Appendix C (Table 6), where we also justify our preferred baseline specification.

Beyond the youth share, Table 2 reveals a negative association between GDP growth and the probability of democratic improvement. This pattern is consistent both with the predictions of our model and with previous empirical evidence (Brückner and Ciccone, 2011; Burke and Leigh, 2010). The lagged Polity level is also negatively associated with democratic improvements, reflecting that countries closer to the upper bound of the Polity scale have less scope for further democratic gains. Finally, the coefficient on GDP per capita is negative but statistically insignificant, consistent with the findings of Acemoglu et al. (2008).

To assess the incremental explanatory power associated with demographic structure, we compare the within- R^2 of nested specifications estimated on a common sample. Using as a starting point the baseline specification of column 5, but omitting both GDP growth and the youth share, adding GDP growth increases the within- R^2 by approximately 10%, while adding the youth share increases it by about 7%. Importantly, when GDP growth is already included, adding the youth share further increases the within- R^2 by roughly 7.5%, a contribution very similar to that obtained when adding the youth share directly. This suggests that demographic structure provides explanatory power beyond the traditional economic determinants of democratic transitions emphasized by AR. Table 13 (Online Appendix O.3) reports the corresponding within- R^2 statistics.

In Online Appendix O.5, we perform sub-sample analyses excluding continents and decades one at a time. The coefficient on the youth share remains relatively stable across regions and periods. It becomes statistically insignificant (at the 5% level) when excluding Africa or Central America, and when excluding the 1970s. However, the point estimates remain positive and sizable, suggesting that the relationship is somewhat stronger in these regions and periods.

Magnitude and economic significance. The estimated effect of the youth share is economically meaningful. In our baseline specification (column 5), a one within-country standard deviation increase in the youth share (0.02) raises the annual probability of democratic improvement by 1.47 percentage points. Relative to an unconditional annual improvement probability of 4.99%, this corresponds to an increase of roughly 30%.¹⁴

To provide a more interpretable benchmark, we evaluate predicted probabilities at different points of each country's within-sample distribution of the youth share. Specifically, for each country we compute the 10th and 90th percentiles of the youth share over the sample period and use the estimated model to predict the probability of democratic improvement when the youth share is set to these values, holding all other covariates at their observed levels and keeping country and year fixed effects unchanged.¹⁵ This yields an average annual probability of 3.40% at the 10th percentile and 6.30% at the 90th percentile, a difference of 2.90 percentage points.

Over a 20-year horizon, under the illustrative assumption that this annual transition probability p remains constant over time, the cumulative likelihood of observing at least one democratic improvement,

¹⁴ In our estimation sample (1950–2018), this benchmark corresponds to variation that occurs frequently within countries: 28.4% of country-year observations exhibit a youth share at least one within-country standard deviation away from the country mean, 84.3% of countries experience such deviations at least once, and the median country experiences them in 22.3% of years.

¹⁵ We first obtain predicted probabilities from the estimated linear probability model with country and year fixed effects and baseline controls. We then construct counterfactual predictions by replacing the observed youth share with the country-specific 10th (or 90th) percentile while keeping all other covariates, as well as the fixed effects, unchanged. The reported probabilities correspond to averages across the estimation sample.

Table 2
Effect of youth bulges on democratic improvements — Regression with controls.

	Binary DV: $\Delta \text{Polity2} \geq 1$					
	Ordinary least squares					
	(1)	(2)	(3)	(4)	(5)	(6)
Youth share (15–19) _{t-1}	0.478*** (0.16)	0.390** (0.18)	0.516*** (0.18)	0.565*** (0.19)	0.490** (0.20)	0.696 (0.50)
ln(GDP p.c.) _{t-1}		–0.012 (0.01)			–0.016 (0.01)	–0.013 (0.03)
Growth rate (GDP p.c.) _{t-1}			–0.002*** (0.00)		–0.002*** (0.00)	–0.003*** (0.00)
Polity _{t-1}				–0.005*** (0.00)	–0.005*** (0.00)	–0.013*** (0.00)
Ceiling control	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	Yes
Country & year FE's	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9030	8613	8588	9030	8587	3269
Countries	163	154	154	163	154	111

Notes: Robust standard errors clustered by country are in parentheses. The unbalanced panel runs from 1950–2018 in the longest specification. Ceiling control equals one if Polity_{t-1} = 10. Additional controls include Urbanization rate, Population (log), Average years of schooling attained by the population aged 15 and above, Gini coefficient, and the Share of employment in the manufacturing sector. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

$1 - (1 - p)^{20}$, increases from about 50% to about 73%. In timing terms, the expected waiting time to the first improvement falls from roughly 29 years to about 16 years (median waiting time from 20 to 10.6 years). These calculations should be interpreted as a benchmark, since they rely on the strong assumption that the annual transition probability remains constant over time.

For comparison, during the Third Wave of democratization (1974–1995) (Huntington, 1993), the annual probability of democratic improvement in our data was 7.0%, compared to 4.0% outside this period—a difference of 3.1 percentage points. Moving from the 10th to the 90th percentile of the youth share within a country increases the annual probability of democratic improvement by approximately 2.9 percentage points, suggesting that within-country demographic variation can generate effects of a similar order of magnitude to major historical episodes of democratic improvements (as a descriptive benchmark).

Are the youngest cohorts driving the effect? Our theoretical framework predicts that revolutionary pressure should be strongest among the youngest cohorts, who face lower opportunity and disutility costs of mobilization and have longer political horizons. We therefore examine whether the estimated effect varies across alternative definitions of the youth population.

Fig. 5(a) reports the coefficient of the youth share from our baseline specification (column 5 of Table 2) using alternative age ranges (15–34, 15–29, 15–24, and 15–19). Fig. 5(b) decomposes the 15–34 group by jointly including the shares aged 15–19, 20–24, 25–29, and 30–34 in the same regression.¹⁶

Fig. 5(a) shows that the estimated coefficient is largest for the 15–19 cohort and becomes smaller and statistically insignificant when broader definitions of youth are used. When all age groups are included simultaneously (Fig. 5(b)), only the share aged 15–19 remains statistically different from zero. Overall, these patterns suggest that the relationship between youth shares and democratic improvements is primarily driven by the youngest adults, consistent with lower opportunity and disutility costs of participation for this cohort.

¹⁶ We define the youth share as the share of the youth cohort (alternatively 15–34, 15–29, 15–24, 15–19, 20–24, 25–29, or 30–34) in the population aged 15 and above. Some studies instead define youth bulges relative to total population; Urdal (2006) discusses shortcomings of that measure and recommends the share of youth within the adult population, which we adopt. This measure is also more consistent with our theoretical framework.

5.2. Two-stage least squares results

Instrumental variable strategy. We now turn to instrumental variable estimates that exploit variation in the youth share driven by lagged fertility dynamics, as described in Section 4. We consider two instruments: domestic fertility and fertility in neighboring countries, both lagged to match the cohort structure of the youth population.

These two instruments provide complementary sources of variation. Domestic fertility offers a strong predictor of cohort size, while neighboring fertility relies on more external variation through the diffusion of fertility behaviors across countries.

Because these instruments rely on long lags, their use mechanically restricts the sample to a shorter panel. To facilitate comparison between OLS and IV strategies, Table 3 reports, for each instrument, OLS estimates computed on the same restricted samples used in the IV specifications. These estimates allow us to compare baseline and IV results within a common sample and to assess the relationship between the instruments and democratic improvement. All specifications include our standard set of controls as well as country and year fixed effects.

Results. Panel A of Table 3 reports the main estimates. Columns 1–3 correspond to the sample used for the IV specification that instruments the youth share with the country's own fertility lagged by 16 years. Column 1 reports the OLS association between the youth share and democratic improvement in this restricted sample. Column 2 presents the reduced-form relationship between lagged fertility and democratic improvement. Column 3 reports the second-stage IV estimate in which the youth share is instrumented by lagged fertility. The IV coefficient is positive and statistically significant, indicating that increases in the youth share raise the probability of democratic improvement.

Columns 4–8 report results for the specifications relying on neighboring countries' fertility lagged by 21 years. Columns 4 and 5 provide OLS estimates in the corresponding restricted sample: column 4 reports the association between the youth share and democratic improvement, while column 5 reports the association between domestic fertility and democratic improvement. Column 6 presents the reduced-form effect of neighboring fertility on democratic improvement.

We then exploit the idea that fertility dynamics diffuse across borders: higher fertility in neighboring countries can influence domestic fertility behavior few years later, which in turn affects the size of domestic youth cohorts. Accordingly, we instrument both the youth share and the domestic fertility rate using lagged neighboring fertility. Columns 7 and 8 report the resulting second-stage estimates. The instrumented demographic variables both have a positive and statistically significant effect on the probability of democratic improvement.

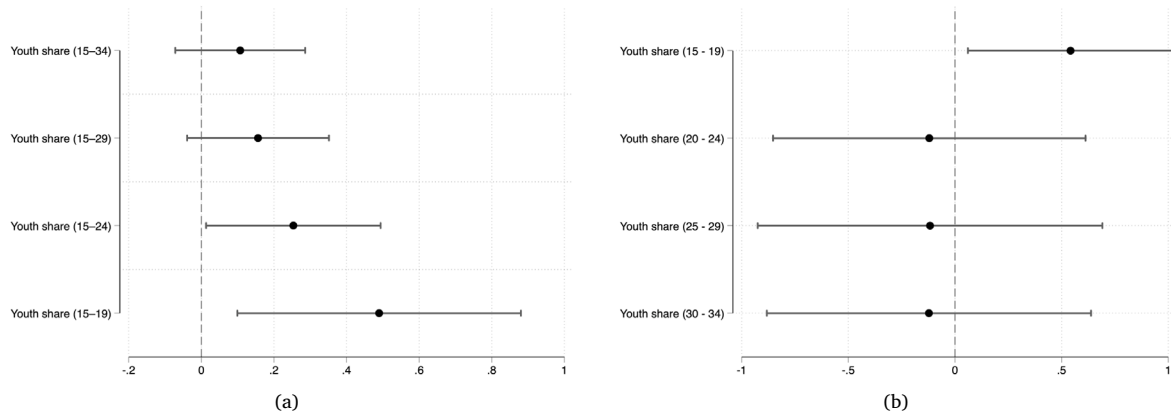


Fig. 5. Effect of youth bulges on democratic improvements — Decomposition by age groups.

Notes: Figs. 5(a) and 5(b) report the coefficient of the youth share from OLS regressions including baseline controls (lagged values of the log of GDP per capita, the growth rate of GDP per capita, and the Polity score). Age cohorts are included either separately (Fig. 5(a)) or jointly in the same regression (Fig. 5(b)). Horizontal bars represent 95% confidence intervals.

Table 3

Effect of youth bulges on democratic improvements — Lagged fertility variables as instruments.

Panel A: Binary DV: Δ Polity2 ≥ 1								
	IV: Own net fertility rate (lagged 16 years)			IV: Neighbors' net fertility rate (lagged 21 years)				
	OLS		2SLS	OLS			2SLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Youth share (15–19) _{t-1}	0.752*** (0.25)		0.798*** (0.28)	0.870*** (0.29)			1.757*** (0.70)	
Net fertility rate _{t-16}		0.016*** (0.01)			0.018*** (0.01)			0.037*** (0.01)
Neighbors' net fertility _{t-21}						0.027*** (0.01)		
Panel B: First stage of 2SLS								
			(3)				(7)	(8)
Net fertility rate _{t-16}			0.02*** (0.00)					
Neighbors' net fertility _{t-21}							0.016*** (0.00)	0.748*** (0.09)
Ceiling control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country & year FE's	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
K-P F-stat on excl. IV's	—	—	817.97	—	—	—	61.25	67.15
Observations	6826	6826	6826	5726	5726	5726	5726	5726
Countries	154	154	154	152	152	152	152	152

Notes: Robust standard errors clustered by country are in parentheses. Ceiling control equals one if Polity2_{t-1} = 10. The standard set of controls includes once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score. Referring to the Kleibergen-Paap (K-P) F-statistic, the test's null hypothesis is that the set of instruments is weak. The panel runs from 1970–2018 for the own net fertility rate specification and from 1975–2018 for the neighborhood net fertility rate specification. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

Panel B reports the first-stage results of the 2SLS procedure. The instruments display strong predictive power. The Kleibergen–Paap F-statistics are far above the conventional rule-of-thumb threshold of 10, reaching nearly 820 for the own-fertility instrument and exceeding 60 for the neighboring-fertility instrument. These results alleviate concerns about weak instruments. Moreover, the first-stage estimates provide evidence consistent with fertility spillovers across countries. Higher domestic fertility strongly predicts a larger youth cohort sixteen years later, while higher fertility in neighboring countries predicts increases in domestic fertility and, ultimately, the youth share, consistent with cross-border diffusion in fertility behavior.

Finally, we compare IV and OLS estimates. While the IV coefficients are generally larger than the baseline OLS estimates reported in Table 2, part of this difference reflects changes in sample composition induced

by the use of long lags. When comparing estimates within the same restricted samples (columns 1 and 3, columns 4 and 7, and columns 5 and 8), the gap between OLS and IV coefficients is substantially reduced. With the own-fertility instrument, the IV estimate is very close to the corresponding OLS estimate. With the neighboring-fertility instrument, the IV coefficient remains larger, although of the same order of magnitude.

This pattern is consistent with the presence of unobserved factors that may affect both demographic structure and democratic outcomes, potentially biasing OLS estimates. For instance, some factors associated with demographic transition may simultaneously foster democratic change while reducing fertility, leading to a downward bias in OLS

estimates.¹⁷ Alternatively, the IV estimates — particularly those based on regional fertility — may capture a local average treatment effect associated with demographic shocks driven by fertility dynamics, which could differ from the average relationship in the full sample.

To further assess this interpretation, we examine how OLS, first-stage, and IV estimates vary when excluding specific continents and time periods (see the Figures reported in Online Appendix O.5). Across continents, excluding Africa reduces OLS, first-stage, and IV coefficients in a similar way, indicating that the stronger contribution of this region reflects underlying heterogeneity in the relationship rather than a distinct subset of observations driving the IV estimates. Across decades, OLS and first-stage coefficients remain relatively stable, while IV estimates display somewhat greater sensitivity to the exclusion of particular periods, suggesting that the IV may place relatively more weight on specific margins of variation. Taken together, these results provide only limited and suggestive evidence of local treatment effects, and do not point to heterogeneity as the primary driver of the difference between OLS and IV estimates.

Overall, the comparison between OLS and IV estimates suggests that the positive relationship between youth cohorts and democratic improvement is robust across specifications, while highlighting the importance of interpreting IV estimates with caution.

Assessing the relevance and validity of the instruments. Our identification strategy relies on lagged fertility as a source of variation in subsequent youth cohort size. We assess the relevance and plausibility of the instruments through a series of complementary exercises examining (i) the mapping between fertility and cohort size, (ii) the timing implied by the cohort mechanism, (iii) potential regional confounding affecting the neighboring-fertility instrument, and (iv) robustness to weak instruments and influential observations.

Mapping fertility into cohort size. A first concern is whether past fertility maps cleanly onto the size of youth cohorts fifteen years later. In principle, this relationship may be blurred by child mortality or migration flows, and fertility itself may respond to economic, political, or social conditions that also affect democratic outcomes.

We address these issues in two ways. First, we refine the fertility-based instruments by adjusting fertility for the probability of surviving from childhood to adolescence and by controlling for average net migration between the date of fertility realization and the date at which cohort size is measured (Table 4, columns 1 and 2). Second, we condition on economic and political characteristics measured at the time fertility decisions were made by adding lagged values of GDP per capita, GDP growth, and the Polity score measured sixteen years earlier, as well as additional lagged controls such as conflict incidence and urbanization (Appendix Table 7).

Across all specifications, the estimated effect of the youth share on democratic improvement remains positive, statistically significant, and of similar magnitude to our baseline results. This stability holds for both instruments and is preserved when we account for survival, migration, and lagged economic and political conditions. Moreover, lagged fertility remains a strong predictor of subsequent cohort size once these intermediate demographic adjustments are taken into account. In fact, adjusting fertility for survival probabilities and controlling for migration modestly strengthens the first stage, as reflected in the Kleibergen–Paap F-statistics reported in Panel B of Table 4. These findings support the relevance of our instruments and are consistent

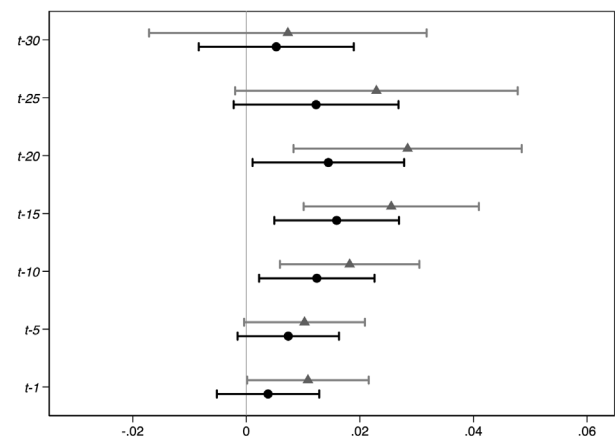


Fig. 6. Timing test for the fertility-based instruments.

Notes: The figure reports reduced-form coefficients from regressions of democratic improvements on different lags of fertility. Black circles correspond to domestic fertility and gray triangles to neighboring countries' fertility. All specifications include country and year fixed effects as well as the baseline set of controls (once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score). The horizontal bars represent 95% confidence intervals.

with the interpretation that past fertility fluctuations translate into predictable variation in the youth share.

Timing and cohort mechanism. We next examine whether the timing of the reduced-form relationship is consistent with the demographic mechanism underlying the instruments. If fertility affects political outcomes only through its impact on cohort size, the effect should emerge only once the relevant birth cohorts reach adolescence. Fertility realizations occurring at substantially shorter or longer horizons should therefore not influence democratic improvements.

We test this implication by estimating reduced-form regressions using alternative lags of fertility. Fig. 6 plots the estimated coefficients for different lags of both domestic fertility (black circles) and neighboring countries' fertility (gray triangles). Consistent with the demographic interpretation of our identification strategy, the estimated effects are concentrated around the lag structure used in our baseline specification, while shorter and longer lags generally produce smaller and statistically indistinguishable effects.

In particular, the effect of domestic fertility peaks around the $t - 15$ lag, which corresponds closely to the horizon at which fertility shocks translate into variation in the youth population. For neighboring countries' fertility, the estimated effect peaks around the $t - 20$ lag, consistent with our assumption that fertility changes abroad affect domestic fertility with a short delay before influencing the size of youth cohorts fifteen years later. One partial exception is the fertility lag at $t - 10$, which remains positive and statistically significant, likely reflecting the persistence of fertility dynamics and the resulting overlap between adjacent cohorts. Overall, the timing pattern is consistent with the view that the relationship between fertility and democratic improvements operates primarily through the cohort-size channel.

Regional confounding and neighboring fertility. A more specific concern relates to the exclusion restriction for the neighboring-fertility instrument. Regional fertility patterns may be shaped by common economic, political, or cultural shocks that also affect democratization dynamics. In addition, democratization itself may display spatial spillovers, so that transitions in neighboring countries influence the probability of transition in country i (Betz et al., 2020).

A natural response would be to introduce region-specific time trends. In practice, however, these trends absorb most of the slow-moving regional variation underlying the instrument. Consistent with this, when region-specific trends are included the first stage collapses:

¹⁷ For instance, the development process is associated with increases in women's education, empowerment, and urbanization. These factors tend to reduce fertility while also fostering democratic change, for example by strengthening social demands for political participation and facilitating citizen coordination. Since these channels are only imperfectly captured by GDP and are likely highly non-linear in development, OLS estimates may be biased downward.

Table 4

IV estimation with child mortality, migration and neighbors' control variables.

Panel A: Binary DV: Δ Polity2 ≥ 1				
	IV: Own fertility	IV: Neighbors' fertility		
	(1)	(2)	(3)	(4)
Youth share (15–19) _{<i>t-1</i>}	0.996*** (0.32)	1.487** (0.68)	1.307* (0.68)	1.696** (0.78)
Average net migration _{<i>t-16</i>→<i>t-1</i>}	-0.161 (0.10)	-0.141 (0.11)		
Neighbors' transition _{<i>t-1</i>}			0.241*** (0.06)	
Neighbors' ln(GDP p.c.) _{<i>t-1</i>}				-0.01 (0.02)
Neighbors' growth rate(GDP p.c.) _{<i>t-1</i>}				-0.002 (0.00)
Neighbors' Polity _{<i>t-1</i>}				0.002 (0.00)
Panel B: First stage of 2SLS				
	(1)	(2)	(3)	(4)
Adjusted net fertility rate _{<i>t-16</i>}	0.02*** (0.00)			
Adjusted neighbors' net fertility _{<i>t-21</i>}		0.016*** (0.00)		
Average net migration _{<i>t-16</i>→<i>t-1</i>}	-0.052*** (0.02)	-0.021 (0.02)		
Neighbors' net fertility _{<i>t-21</i>}			0.015*** (0.00)	0.015*** (0.00)
Ceiling control	Yes	Yes	Yes	Yes
Standard controls	Yes	Yes	Yes	Yes
Country & year FE's	Yes	Yes	Yes	Yes
K-P F-stat on excl. IV's	884.10	76.36	58.42	51.73
Observations	6826	5872	5366	5135
Countries	154	152	153	153

Notes: Robust standard errors clustered by country are in parentheses. Ceiling control equals one if Polity2_{*t-1*} = 10. The standard set of controls includes once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score. Referring to the Kleibergen-Paap (K-P) F-statistic, the test's null hypothesis is that the set of instruments is weak. The panel runs from 1970–2018 for the own net fertility rate specification and from 1975–2018 for the neighborhood net fertility rate specification. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

the coefficient on neighboring fertility becomes statistically insignificant and the Kleibergen–Paap F-statistic falls to 0.39. We therefore adopt a more targeted strategy. First, following Betz et al. (2020), we control for democratic improvements in neighboring countries at time $t - 1$. The results are reported in column 3 of Table 4. Neighboring democratic improvements positively affect the probability of democratization in country i , but the estimated effect of the youth share remains positive and statistically significant. Its magnitude decreases by roughly 25%, suggesting that regional spillovers may account for part of the baseline estimate. However, including democratic improvements in neighboring countries as a control raises additional concerns. Political transitions may spill over across countries, and common regional shocks may simultaneously affect both domestic and neighboring outcomes. As a result, this variable may not be a standard control and could introduce additional biases, which makes its interpretation less straightforward. For this reason, we treat specifications including this control as robustness checks and do not include it in our baseline regressions.

Second, we control for regional averages of GDP per capita, GDP growth, and Polity scores (column 4 of Table 4). Reassuringly, the estimated effect of the youth share remains positive and statistically significant when these regional controls are included. In Appendix Table 8, we further combine these regional controls with additional domestic variables capturing plausible alternative channels through which regional demographic shocks may affect democratization, namely conflict incidence, migration, and urbanization. As these variables are measured at $t - 1$, they may partly reflect intermediate channels rather than pure confounders. Including them therefore provides a stringent robustness check. Reassuringly, the estimated effect of the youth share remains positive and statistically significant across specifications, while the first-stage relationship remains sufficiently strong.

Placebo outcomes and development channels. As an additional check on the exclusion restriction, we examine whether the instruments predict alternative socioeconomic outcomes that could reflect broader development dynamics rather than cohort-size variation. Specifically, Table 9 reports reduced-form regressions of urbanization rates and years of schooling on our baseline instruments.

The results show that neither instrument significantly predicts urbanization. Lagged domestic fertility is negatively associated with schooling outcomes, while the neighboring-fertility instrument does not significantly affect educational attainment. Given the long lag between fertility realizations and schooling outcomes, this pattern is consistent with the quantity–quality trade-off in child investment, whereby smaller cohorts are associated with higher educational attainment.

Importantly, our main IV estimates remain stable when controlling for schooling in the second-stage regressions (Table 9, columns 5 and 6). This suggests that the relationship between the youth share and democratic improvements is not driven solely by observable development channels. Moreover, the absence of a significant relationship between the neighboring-fertility instrument and schooling outcomes is consistent with the idea that this instrument is less directly linked to domestic socioeconomic dynamics.

Finally, as an additional and suggestive check, we examine whether the instruments retain predictive power for democratic improvements once controlling for the youth share. We find that the reduced-form relationship disappears when the youth share is included as a control (see Table 9, columns 7 and 8), which is consistent with the interpretation that the effect of the instruments operates primarily through cohort-size variation rather than alternative development channels.

Weak instruments and influential observations. A final concern relates to the potential weakness of the neighboring-fertility instrument. While

the Kleibergen–Paap F-statistics reported in Tables 3–8 are generally above conventional thresholds, we complement these diagnostics with weak-instrument-robust inference. Anderson–Rubin confidence intervals (Online Appendix Table 21) confirm that the estimated effects remain statistically significant. We also assess the sensitivity of our results to influential observations (Young, 2022) using leave-one-country-out (jackknife) estimations. The results remain stable and statistically significant throughout (Online Appendix Figure 21), indicating that they are not driven by a small number of countries.

Taken together, these exercises provide consistent evidence in favor of the relevance of our instruments and support the interpretation that fertility-based variation affects democratic improvement primarily through the cohort-size channel. At the same time, as in most applications relying on historical instruments, the exclusion restriction cannot be fully tested and the results should therefore be interpreted with appropriate caution.

5.3. Evidence based on demographic transition timing

We next implement the alternative empirical strategy based on the timing of demographic transitions. Building on the approach described in Section 4, we exploit cross-country variation in the timing of fertility peaks to identify periods during which countries are expected to experience relatively large youth cohorts. Specifically, for each country, we define a reference year corresponding to the fertility peak shifted forward by 17 years, which approximates the period during which cohorts born at peak fertility reach youth ages. We then construct time windows around this reference date and examine how the probability of democratic improvement varies within these windows.

We begin by estimating a set of specifications based on symmetric windows of increasing size around the transition date (e.g., ± 5 , ± 10 , ± 15 years). In each case, we include a dummy variable equal to one if a country-year observation falls within the corresponding window, and zero otherwise. These regressions include country and year fixed effects as well as our baseline set of controls. This approach allows us to assess whether democratic improvements are more likely to occur in periods close to demographic transitions, and whether this relationship strengthens as the window expands.

We then refine this analysis by decomposing time into disjoint intervals relative to the transition date (e.g., $[-25, -15]$, $[-15, -5]$, $[-5, 5]$, $[5, 15]$, $[15, 25]$). This specification provides a more direct characterization of the temporal structure of the relationship between demographic transitions and democratic improvements by comparing the probability of transition across periods before, around, and after the emergence of large youth cohorts. Because the omitted category consists of observations sufficiently far from the transition date, the estimated coefficients can be interpreted as deviations from baseline periods in which demographic pressures are expected to be weaker.

An important advantage of this approach is that it can be implemented over a longer historical period. Since it relies only on the timing of fertility transitions rather than detailed demographic data, we are able to extend the sample back to the nineteenth century for a subset of countries. This provides additional variation and allows us to assess whether the relationship between demographic transitions and democratic improvement holds over a longer time horizon.

Fig. 7 reports the corresponding results.¹⁸ Panel (a) shows estimates obtained from a series of regressions in which we include, one at a time, indicators for symmetric windows of increasing size around the transition date. A clear monotonic pattern emerges: the estimated effect decreases with the size of the window. This indicates that democratic improvements are increasingly concentrated as observations move closer to the transition period, and that the relationship

strengthens as the window captures a larger share of the transition phase.

Panel (b) reports estimates from a single regression that includes indicators for multiple disjoint intervals relative to the transition date. It provides a more direct view of the temporal structure of the relationship. The probability of democratic improvement is highest in the period immediately surrounding the transition date, while it is lower both in earlier and later periods. This pattern is consistent with a temporary effect of demographic change, concentrated around the emergence of large youth cohorts.

These results provide complementary evidence that democratic improvements are not evenly distributed over time, but instead cluster around periods associated with demographic transitions. While these estimates should be interpreted as descriptive rather than causal, they reinforce the interpretation that demographic structure plays an important role in shaping the likelihood of democratic improvement.

6. Further analysis

6.1. Recessions, youth bulges, and democratic improvements

According to our theoretical framework, the effect of economic downturns on democratic improvements should be amplified when the youth share is high. To test this prediction, we extend our baseline linear probability model by interacting the youth share with different measures of economic conditions. We first consider GDP per capita growth, then a binary recession indicator equal to one when growth is negative, and finally a truncated negative growth rate that captures the severity of economic downturns.

The results of these specifications, reported in Table 14 (Online Appendix O.4), are in line with this prediction. The interaction between GDP growth and the youth share is negative, while the interactions involving the recession indicator and the truncated negative growth rate are positive. Fig. 8(a) illustrates this result by plotting the marginal effect of a recession (binary measure) on the probability of democratic improvement as a function of the youth share. The estimated effect is small and only weakly significant at low levels of the youth share, but becomes larger and statistically significant when the youth share is high. This pattern indicates that recessions are more likely to trigger democratic improvements in demographic contexts characterized by relatively large youth cohorts.¹⁹

We also examine whether this complementarity between economic and demographic conditions is present in the longer-run panel, using the alternative strategy based on the timing of demographic transitions. In this setting, we define periods of relatively large youth cohorts using symmetric time windows around the fertility peak shifted forward by 17 years, as described above. To assess how the marginal effect of recessions varies across these periods, we estimate separate regressions for observations falling inside and outside such windows. Rather than relying on a single definition of the demographic transition period, we consider several alternative windows, ranging from $[-5, 5]$ to $[-15, 15]$ years around the transition date.

Fig. 8(b) reports the corresponding marginal effects of recessions. A consistent pattern emerges. Outside demographic transition periods, recessions have a positive but relatively modest effect on the probability of democratic improvement. By contrast, within periods associated with large youth cohorts, the effect of recessions is systematically larger.²⁰ This difference remains visible across alternative window definitions, indicating that the result is not driven by a particular choice of timing.

¹⁸ Detailed regression results corresponding to Fig. 7 are reported in Appendix Table 10.

¹⁹ The same pattern emerges when we consider the growth rate of GDP and the truncated negative growth rate of GDP (see Figure 15 in Online Appendix O.4).

²⁰ The same pattern emerges when we consider the growth rate of GDP and the truncated negative growth rate of GDP (see Figure 16 in Online Appendix O.4).

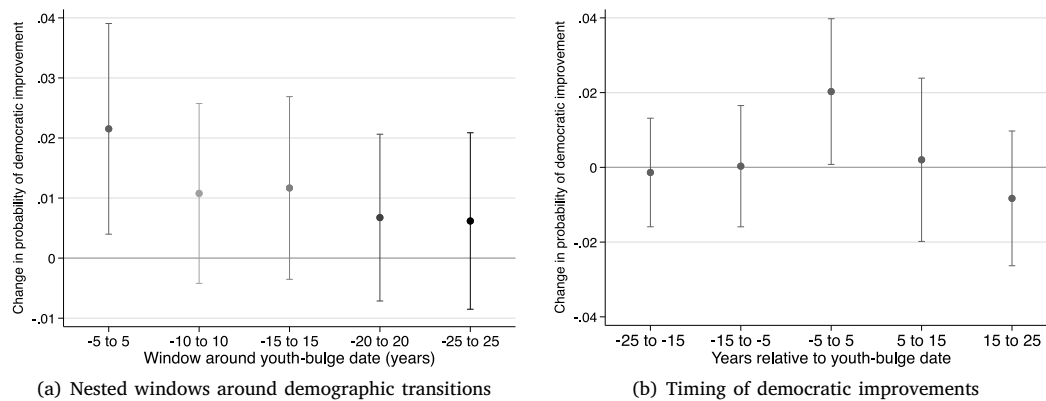


Fig. 7. Democratic improvements around demographic transitions.

Notes: The figures report estimates from linear probability models of democratic improvement on indicators for periods relative to the timing of demographic transitions. For each country, the transition date is defined as the year of peak fertility shifted forward by 17 years to approximate the period when cohorts born at peak fertility reach youth ages. The sample covers the period 1850–2018, depending on data availability across countries. All specifications include country and year fixed effects as well as the baseline set of control variables. Panel (a) reports coefficients from separate regressions in which a single indicator for belonging to a symmetric window around the transition date (of varying size) is included. Panel (b) reports coefficients from a single regression in which multiple indicators for disjoint time intervals relative to the transition date are included simultaneously. The omitted category corresponds to country-years sufficiently far from the transition. All specifications include country and year fixed effects as well as the baseline set of controls (once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score). Confidence intervals are shown at the 90% level.

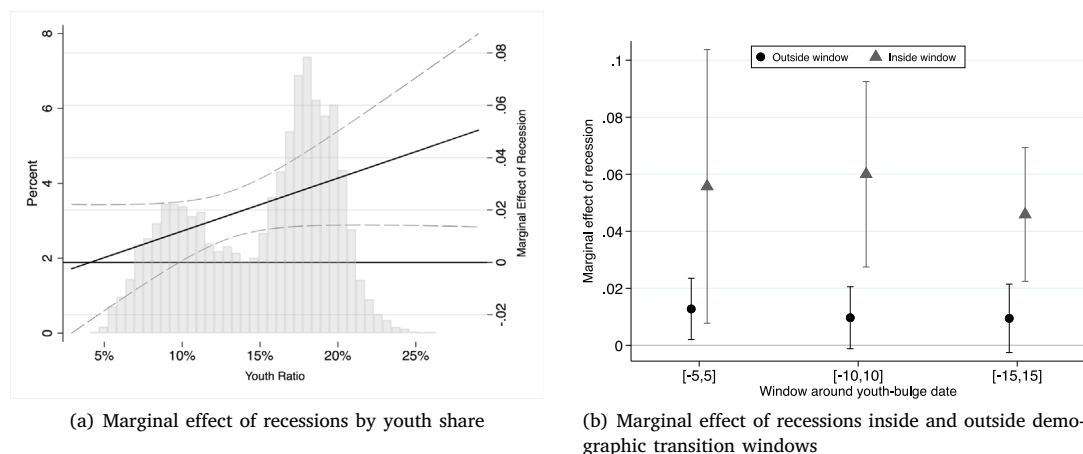


Fig. 8. Interaction effects between recessions and demographic structure.

Notes: Panel (a) plots the marginal effect of a recession on the probability of democratic improvement as a function of the youth share, based on the interaction specification reported in Online Appendix Table 14. The histogram in the background shows the distribution of the youth share. Panel (b) reports the estimated marginal effect of recessions from separate fixed-effects regressions run inside and outside symmetric windows around the fertility peak shifted forward by 17 years, which approximates the period when cohorts born at peak fertility reach youth ages. In both panels a recession is a dummy variable equals to one of the GDP growth rate by capita is negative. The sample in panel (b) covers the period 1850–2018, depending on data availability across countries. All specifications include country and year fixed effects as well as the baseline set of controls (once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score). Confidence intervals are shown at the 95% level.

6.2. Longevity, demographic structure, and democratic improvements

The model features survival probabilities across cohorts, captured by the parameter π , which jointly determine individuals' time horizons and the demographic structure of the population. While the baseline empirical analysis has focused on variation in the youth share, this section examines more directly the role of survival probabilities and their implications for democratic improvements.

The model predicts an ambiguous effect of survival. On the one hand, higher survival probabilities increase the expected duration over which individuals benefit from a regime change, thereby strengthening incentives to support democratization (a “horizon effect”). On the other hand, improvements in survival — particularly at older ages — alter the demographic structure by increasing the relative weight of

older cohorts and reducing the share of younger individuals, thereby weakening revolutionary pressure (a “composition effect”).

To assess these mechanisms empirically, we augment our baseline specifications by including measures of survival probabilities at different ages. While the model captures survival through a single parameter π , which reflects the probability of reaching older ages, the empirical analysis allows us to distinguish between survival at different stages of the life cycle. This provides a way to examine whether the effect of survival depends on the segment of the age distribution that is most affected.

We begin by examining the relationship between survival probabilities and the youth share. We estimate regressions of the youth share on lagged fertility, survival probabilities, and standard controls, with country and year fixed effects. Consistent with the composition

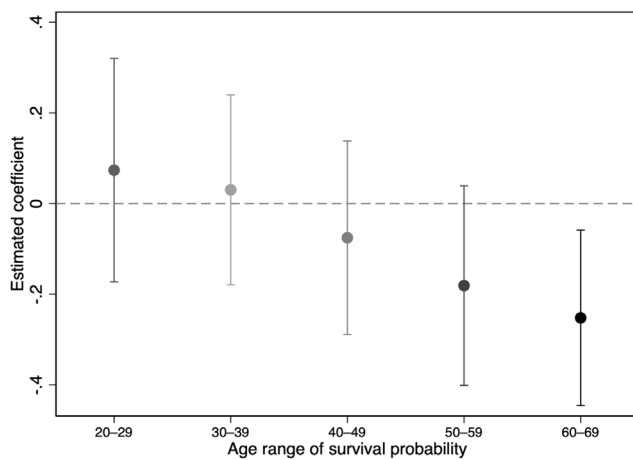


Fig. 9. Survival probabilities and democratic improvement.

Notes: The figure plots the estimated coefficients on lagged survival probabilities from separate OLS regressions of democratic improvement on the youth share. Each point corresponds to a distinct regression including one survival probability measure only. All specifications include country and year fixed effects as well as the baseline set of controls (once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score). Confidence intervals are shown at the 95% level.

effect, we find that higher survival at older ages is associated with a significantly lower youth share, conditional on fertility and other controls (see Online Appendix Table 16).

We then turn to the relationship between survival probabilities and democratic improvements. Fig. 9 plots the coefficients associated with survival probabilities at different ages in otherwise identical OLS specifications. A clear pattern emerges: the estimated effect is close to zero, and even slightly positive, for survival at younger ages, but becomes increasingly negative as survival extends to older age groups. The coefficient is largest in absolute value, and statistically significant, for survival between ages 60 and 69. The full set of regression estimates is reported in Online Appendix Table 17. Importantly, the estimated effect of the youth share remains broadly stable across specifications.²¹

At first glance, this result may appear surprising. Conditional on the youth share, one might expect the coefficient on survival probabilities to primarily capture the horizon effect, which predicts a positive impact on democratic improvement. Instead, we find that the effect becomes increasingly negative as survival extends to older age groups.

A natural interpretation is that survival probabilities continue to capture part of the composition effect even when controlling for the youth share. While the youth share summarizes one dimension of the demographic structure, it does not fully account for the persistence and relative weight of older cohorts. Increases in survival—especially at older ages—therefore still shift the population toward older groups in ways that are not entirely reflected in the youth share, thereby reducing revolutionary pressure.

These findings also suggest that the opportunity cost and disutility channels emphasized in the model play a more central role than the horizon effect in shaping the impact of age structure on democratic improvements. These costs differ sharply across age groups, reflecting discontinuities in economic and social integration, whereas the horizon effect evolves more gradually with age and generates gains that

²¹ We consider survival probabilities over ten-year intervals, from ages 20–29 up to 60–69. Using survival probabilities measured from age 20 onward yields similar qualitative patterns (see Online Appendix Figure 17). This alternative measure more directly captures the horizon effect, but is estimated with less precision due to the longer time span involved.

Table 5

Effect of youth bulges on democratic improvements — Riots as a transmission channel.

	First stage: asinh(riots _{t-1} ^{mm}) (1)	Second stage: Δ Polity2 ≥ 1 (2)
Youth share (15–19) _{t-1}	4.270* (2.19)	
asinh(riots _{t-1} ^{mm}) (instrumented)		0.133* (0.07)
Ceiling control	Yes	Yes
Standard controls	Yes	Yes
Country & year FE's	Yes	Yes
K-P F-stat on excl. IV's	–	3.814
Observations	8234	8234
Countries	154	154

Notes: Robust standard errors clustered by country are in parentheses. The panel runs from 1950–2018 for all specifications. Ceiling control equals one if Polity2_{t-1} = 10. The standard set of controls includes once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score. The variable riots_{t-1}^{mm} denotes the number of riots per million inhabitants in year *t* – 1, asinh(riots_{t-1}^{mm}) is the inverse hyperbolic sine transformation of this measure. The panel runs from 1950–2018. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

are more diffuse and less tightly linked to immediate incentives. The increasingly negative coefficients for higher age groups are consistent with this pattern, suggesting that compositional changes driven by survival are more strongly reflected through the cost channels.

6.3. Riots as a transmission mechanism

We next explore whether riots may constitute a transmission channel between youth bulges and democratic improvement, consistent with the “threat of revolution” hypothesis. In line with our theoretical framework, a large cohort of young individuals may increase the pressure on incumbent regimes by raising the likelihood of collective unrest. While a high youth share may not always be perceived as a credible threat by elites, intermediate events such as riots — i.e., episodes of low-intensity conflict that may escalate — can act as visible signals of underlying social tensions and trigger political concessions (Andrews and Jackman, 2005; Aidt and Franck, 2015; Aidt and Leon, 2016; Parante et al., 2022). In this sense, riots can be interpreted as observable manifestations of a latent revolutionary threat, making it more salient and credible to incumbent elites.

To investigate this mechanism, we use the measure of riots provided by Banks (2020), which counts the annual number of violent demonstrations or clashes involving more than 100 citizens and the use of physical force. Because the distribution of riots is highly skewed and includes many zero values, we use the inverse hyperbolic sine transformation, which accommodates zeroes while reducing the influence of extreme observations. Our baseline specification uses riots per million inhabitants, which allows us to account for differences in country size while preserving meaningful variation across countries.

We proceed in two steps. First, we examine whether youth bulges are associated with increased riot activity. Second, we assess whether the component of riot activity explained by demographic pressure is related to democratic improvement. To this end, we use the youth share to isolate variation in riots driven by demographic factors and examine its relationship with political transitions. This exercise is not meant to identify a causal effect of riots per se, as the exclusion restriction may not fully hold. Rather, it provides a way to assess whether the variation in riots linked to youth bulges is consistent with riots acting as a transmission channel.

Table 5 reports the corresponding estimates. The first stage shows that youth bulges are positively and significantly associated with riot intensity. The second-stage results indicate that the component of riots driven by demographic pressure is positively related to democratic

improvements. However, estimates are relatively imprecise, reflecting the limited variation in riots explained by demographic factors, and first-stage F-statistics are modest. We also consider alternative specifications based on inverse hyperbolic sine transformations of riot counts, with and without controlling for population size. These yield similar qualitative results (see Table 15 in Online Appendix O.4).

The results are overall consistent with the view that riots may serve as one channel through which youth bulges contribute to democratic change, although we interpret this evidence as suggestive rather than conclusive.

7. Robustness

7.1. Alternative instrument: Lagged climatic condition

To further assess the robustness of our results, we consider an alternative instrumental-variable strategy based on past climatic conditions. This approach provides an independent source of variation in fertility dynamics, but relies on a more indirect mechanism and is therefore best interpreted as a robustness exercise rather than as a standalone identification strategy.

Climatic conditions can affect fertility through their impact on household income, particularly in economies that rely heavily on agriculture. We exploit this relationship using the Standardized Precipitation Evapotranspiration Index (SPEI), which combines information on precipitation and soil conditions at a fine geographical level and is aggregated at the country level (see Online Appendix O.6.1). To align the timing with cohort formation, we average SPEI over the period $[t - 17, t - 21]$, corresponding to climatic conditions at the time of birth of individuals aged 15–19 at date t . We interact this measure with the share of agriculture in GDP to capture the idea that climatic shocks have stronger effects in more agriculture-dependent economies.

The first stage shows that positive climatic shocks (higher SPEI values) reduce fertility in countries with a large agricultural sector (Online Appendix Table 18, Panel B), consistent with the idea that a positive income shock increases the opportunity costs of raising children (Becker et al., 1960; Becker and Lewis, 1973; Becker and Tomes, 1976; Rosenzweig, 1977). However, the relationship is weaker and less precisely estimated than in our baseline IV strategy. Second-stage estimates indicate a positive and significant effect of the instrumented youth share on democratic improvements (Online Appendix Table 18, Panel A, column 4), but the IV coefficient is substantially larger than the corresponding OLS estimate. Given the weakness of the first stage and the magnitude of the IV estimates, we interpret these results with caution and treat this strategy as providing only suggestive evidence.

To further assess potential violations of the exclusion restriction, we examine whether the climatic instrument predicts democratic improvements beyond its effect through the youth share. Reduced-form regressions show that the instrument remains significant when controlling for the youth share (Online Appendix Table 18, Panel A), suggesting the presence of additional channels. To mitigate this concern, we augment our baseline specification with controls for regional political and economic conditions, as well as for potential channels such as conflict and migration (Online Appendix Tables 18 and 19). While the 2SLS estimates remain consistent in sign, they are sensitive to specification.²² Finally, we report overidentification tests combining the fertility-based instrument with the climatic instrument (Online Appendix Table 22). Given the concerns discussed above regarding the exclusion restriction for the climatic instrument, the interpretation of these tests is limited, as their validity relies on the joint validity of both instruments.

²² We complement these results with weak-instrument-robust inference and jackknife sensitivity analyses, which yield similar qualitative conclusions (see Online Appendix O.6.5).

7.2. Different measures of political transition

In this section, we assess the robustness of our baseline results to alternative measures of democratic improvement. Online Appendix Figure 19 reports estimates obtained using four different operationalizations of the dependent variable.

Specifically, we consider the following measures of political transition: (i) the change in the Polity2 score, (ii) the truncated change in the Polity2 score (defined as positive changes only, and zero otherwise), (iii) an increase of at least 5% in the Machine Learning (ML) Democracy Index proposed by Gründler and Krieger (2016, 2021), and (iv) an increase of at least 5% in the V-Dem Polyarchy index.²³ For each outcome, we estimate three specifications: a baseline OLS regression and two instrumental variable (IV) regressions, using domestic fertility and neighboring countries' fertility as instruments for the youth bulge. All specifications include the same set of controls and fixed effects as in our baseline analysis.

Overall, the results are consistent across alternative measures of democratic improvement. The estimated coefficients remain positive across all specifications and are statistically significant in most cases, supporting the robustness of our main findings. Importantly, the relative magnitude of the IV estimates remains similar across specifications, with estimates based on neighboring fertility generally slightly larger than those based on domestic fertility.

7.3. Placebo tests using alternative age cohorts

To further support the validity of our identification strategy and address concerns regarding the exclusion restriction, we conduct a placebo exercise using age cohorts that, according to our framework, are unlikely to directly affect political transitions.

Specifically, we replicate our baseline specification while replacing the youth bulge variable (defined as the ratio of individuals aged 15–19 to the population aged above 15) with analogous population ratios constructed for cohorts that are either too young or too old to be politically relevant in our setting, namely ages 0–4 and cohorts aged 30 and above. Online Appendix Figure 20 reports the estimated coefficients associated with these alternative cohort shares. Most coefficients are small in magnitude and statistically indistinguishable from zero. In addition, estimates for older cohorts tend to be negative, although generally imprecisely estimated.

These results suggest that our findings are not driven by general demographic trends, by the fact that fertility changes affect the size of all age cohorts over time, or by persistent institutional or societal characteristics that may jointly influence demographic structure and the likelihood of transition, thereby providing additional support for the exclusion restriction.

7.4. Time varying country fixed effects

As shown on the Online Appendix Table 20, the positive relationship between the youth population share and democratic improvements is robust to controlling for flexible time-varying heterogeneity. In particular, the coefficient remains large and statistically significant when allowing for country-specific linear trends. When replacing country fixed effects with country-by-period fixed effects, the estimates remain positive but become less precise. This might reflect the fact that demographic variables evolve gradually over time, and country-by-period fixed effects absorb much of this low-frequency variation, leaving only short-run deviations within each period to identify the effect.

²³ We adopt a 5% threshold for consistency across measures. The ML and V-Dem indices are defined on a $[0, 1]$ scale, while the Polity2 index ranges from -10 to 10 , so a one-point increase in Polity2 corresponds to a 5% change on its scale.

8. Conclusion

In this paper, we study how demographic structure shapes the likelihood of democratic improvements. Using detailed demographic data in a long country-year panel, we document a robust empirical regularity: the share of individuals aged 15–19 is positively associated with the probability of democratic improvement. An instrumental-variable strategy based on lagged fertility, together with complementary evidence, is consistent with an interpretation in which demographic structure plays a causal role, although identification challenges remain. We also establish several additional results. First, the effect of the youth share is amplified during periods of economic recession. Second, increases in survival probabilities — especially at older ages — are associated with a lower likelihood of democratic improvement, consistent with the compositional effects highlighted in the model. Finally, riots appear to act as one potential transmission channel through which youth bulges translate into political change.

We view this set of results as supportive of the AR window of opportunity theory. As younger cohorts are those who have the most to gain from a regime change and the least to lose from participating in democratic movements, a temporary rise in the youth share induces a threat of revolution (especially during recessions as economic opportunities are more depressed). As a result of this threat, and to preempt large scale civil conflicts, the elite in place are incentivized to concede democratic improvements (especially when riots lend credibility to the perceived threat). While the existing empirical literature has shown the role played by economic variables in the opening of a democratic window of opportunity, our paper demonstrates that demographic variables also play a crucial role and complement the macroeconomic forces. We believe that this result constitutes a useful step toward a more systematic investigation of all the factors conducive to democracy.

CRedit authorship contribution statement

Rédha Chaba: Writing – original draft, Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization, Data curation,. **Michael T. Dorsch:** Writing – original draft, Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Victor Hiller:** Writing – original draft, Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Paul Maarek:** Writing – original draft, Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The model - A role for protest in the transition to democracy

We consider a modified version of the model presented in Section 2. To emphasize the role of protests and their interaction with the youth share, we simplify the demographic structure by assuming that individuals live for only two periods, indexed by $i \in \{y, m\}$. We further assume $\varepsilon = 0$, so that the gross income of a citizen of age i is ω_i .

We normalize the cost of participating in a revolution for mature individuals to zero, $\chi_m = 0$.²⁴ In contrast, the cost of participating in a revolution for young individuals is stochastic and depends on idiosyncratic socio-political circumstances. Specifically, $\chi_y = \chi$, where χ takes value 0 with probability λ and $\hat{\chi} > 0$ with probability $1 - \lambda$. Citizens observe the realization of χ at the beginning of the period, whereas the elite does not.

²⁴ This normalization is without loss of generality, as mature individuals never find it optimal to participate in a revolution in this two-period economy, since they do not benefit from a regime change in a subsequent period.

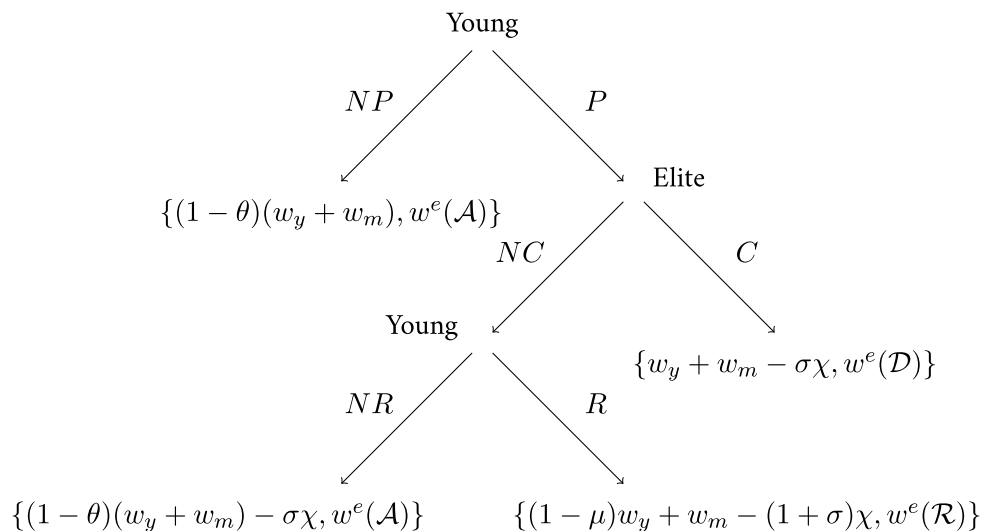


Fig. 10. Decision tree.

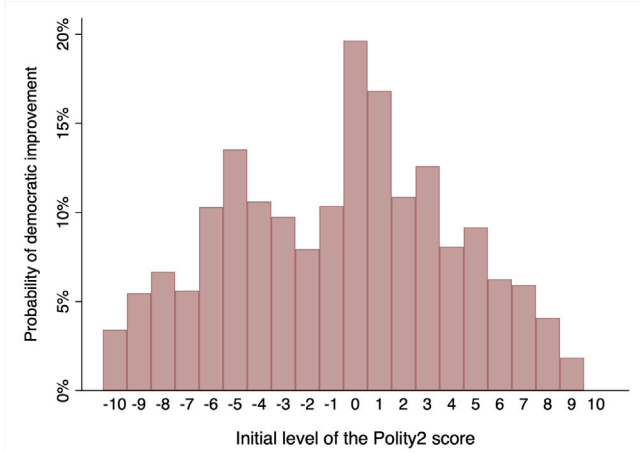


Fig. 11. Probability of democratic improvement by initial level of Democracy.

We introduce an additional stage in the interaction between citizens and the elite. At the beginning of each period t , after observing the realizations of ζ and χ , citizens decide whether to participate in protests against the elite. Participating in protests entails a fixed cost $\sigma\chi$, where $\sigma \in (0, 1)$ captures the degree of correlation between the cost of protests and the cost of revolution. Protests are a necessary prerequisite for launching a revolution later in the period. When deciding whether to concede democracy, the elite observes citizens' protest participation but does not observe the underlying realization of χ .

Fig. 10 summarizes the possible decision sequences and the payoffs associated with each scenario when $\zeta \leq \psi_y$. Before describing the figure, a few preliminary comments are necessary. First, as in the baseline version of the model, μ is assumed to exceed θ , and since citizens live for only two periods, it is never in the interest of mature individuals to participate in revolutionary movements. Therefore, the analysis focuses on the interaction between the two relevant players: the young and the elite.

Citizens move first by deciding whether to protest (P) or not (NP). If they choose not to protest, a revolution cannot occur within the period, and the elite have no incentive to concede democracy. In this

case, the payoffs correspond to those under autocracy: $w^e(\mathcal{A})$ for the elite and $(1 - \theta)(w_y + w_m)$ for the young.²⁵ If citizens choose to protest, the elite may decide to either concede democracy (C) or not (NC). If the elite concedes, the payoffs correspond to those under a democracy, reduced by $\sigma\chi$ (the cost of protesting) for the citizens. If the elite choose not to concede democracy (NC), young citizens face the decision to either revolt (R) or not revolt (NR). If they choose not to revolt, the political regime remains an autocracy, but citizens still incur the protest cost $\sigma\chi$. Conversely, if young citizens decide to revolt, democracy is established, and the resulting payoffs reflect this transition, reduced by both $\sigma\chi$ (the cost of protesting) and χ (the cost of revolting) for the citizens.

Let us assume that the following parameter configurations hold.

Assumption 3.

- (1) $\theta w_m - (\mu - \theta)w_y > 0$
- (2) $\hat{\chi} > \frac{\theta}{\sigma}(w_y + w_m)$
- (3) $\lambda < \frac{u^e(\mathcal{A}) - u^e(D)}{u^e(\mathcal{A}) - u^e(R)}$

We seek the perfect Bayesian Nash equilibrium of this game (in pure strategy). To characterize this equilibrium, we employ backward reasoning. At the final stage, if, at previous stages, the young have chosen to protest and the elite have decided not to concede, the young will choose to revolt if and only if

$$\chi \leq \theta w_m - (\mu - \theta)w_y.$$

Therefore, under Assumption 3(1), young citizens revolt in the state where $\chi = 0$ (denoted as state σ_1), while Assumption 3(2) ensures that they do not revolt in the state where $\chi = \hat{\chi}$ (denoted as state σ_2).

Let $\mu(\sigma_i|P)$ denote the probability that the elite assign to state σ_i after observing action P . At stage 2, the elite decide whether to concede democracy by comparing their utility under democracy, $w^e(D)$, with their expected utility from not conceding. Since, at the final stage, young citizens revolt only in state σ_1 , this expected utility is given by:

$$\mu(\sigma_1|P)w^e(R) + \mu(\sigma_2|P)w^e(\mathcal{A}),$$

²⁵ We continue to assume myopic expectations, meaning that young citizens anticipate the regime will remain autocratic in the subsequent period.

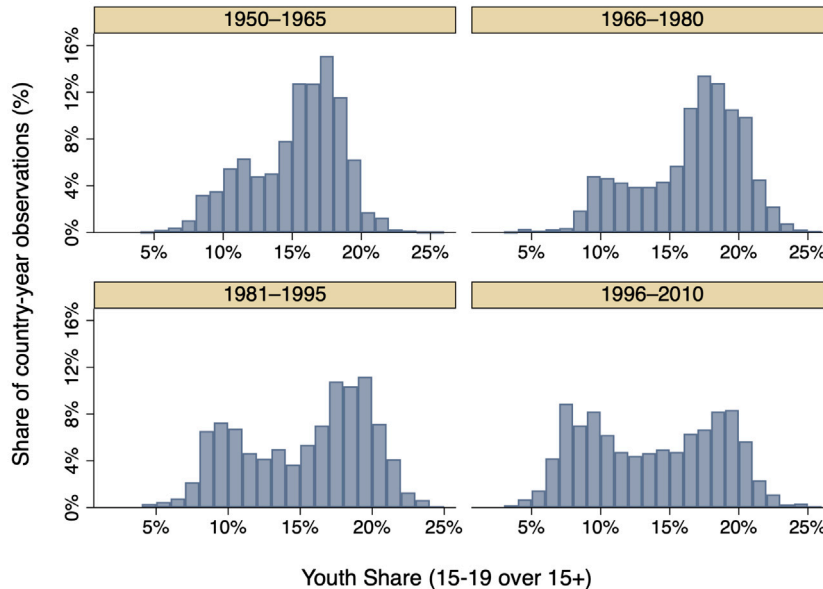


Fig. 12. Evolution of the youth share distribution over time.

Table 6
Effect of youth bulges on democratic improvements — Regression with additional controls.

	Binary DV: $\Delta \text{Polity2} \geq 1$					
	Ordinary least squares					
	(1)	(2)	(3)	(4)	(5)	(6)
Youth share (15–19) _{t-1}	0.479*** (0.16)	0.482*** (0.17)	0.416** (0.19)	0.585* (0.32)	0.631** (0.29)	0.696 (0.50)
Urbanization rate _{t-1}	-0.000 (0.00)					-0.000 (0.00)
ln(population) _{t-1}		-0.001 (0.01)				0.048 (0.06)
Years of schooling _{t-1}			-0.001 (0.00)			0.000 (0.00)
Gini coefficient _{t-1}				-0.001 (0.00)		0.000 (0.00)
% in Manufacturing _{t-1}					0.006** (0.00)	0.007 (0.01)
ln(GDP p.c.) _{t-1}						-0.011 (0.03)
Growth rate (GDP p.c.) _{t-1}						-0.003*** (0.00)
Polity _{t-1}						-0.015*** (0.00)
Ceiling control	Yes	Yes	Yes	Yes	Yes	Yes
Country & year FE's	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9053	9053	7601	4929	5547	3272
Countries	164	164	135	160	140	113

Notes: Robust standard errors clustered by country are in parentheses. The unbalanced panel runs from 1950–2018 in the longest specification. Ceiling control equals one if Polity2_{t-1} = 10. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

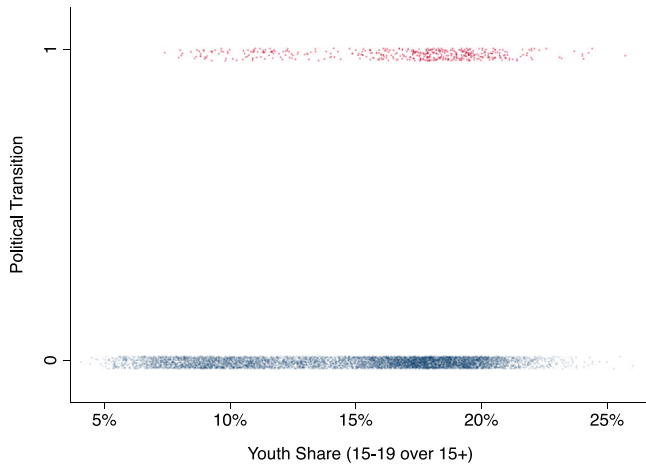


Fig. 13. Youth share and democratic improvement: raw data.
Notes: Each point represents a country-year observation. The vertical axis indicates whether a democratic improvement occurs. To improve readability, a small random perturbation is added to the vertical position of the points.

so that the elite choose to concede democracy if and only if

$$\mu(\sigma_1|P) \geq \frac{w^e(\mathcal{A}) - w^e(D)}{w^e(\mathcal{A}) - w^e(R)}.$$

Applying Bayes' rule, we obtain

$$\mu(\sigma_1|P) = \frac{\mathbb{P}(P|\sigma_1)\mathbb{P}(\sigma_1)}{\mathbb{P}(P|\sigma_1)\mathbb{P}(\sigma_1) + \mathbb{P}(P|\sigma_2)\mathbb{P}(\sigma_2)} = \frac{\lambda}{\lambda + (1 - \lambda)\mathbb{P}(P|\sigma_2)}.$$

The final equality follows from the fact that, in state σ_1 , the young citizens' utility from revolution exceeds their utility from not protesting, implying $\mathbb{P}(P|\sigma_1) = 1$, and by assumption, $\mathbb{P}(\sigma_1) = \lambda$. Since we focus on pure strategies, $\mathbb{P}(P|\sigma_2)$ can only take two values: 0 or 1.

It is straightforward to show that, in equilibrium, $\mathbb{P}(P|\sigma_2)$ cannot be equal to 1. Suppose, by contradiction, that $\mathbb{P}(P|\sigma_2) = 1$, meaning that young citizens always choose to protest in state σ_2 . In this case, $\mu(\sigma_1|P)$

would be equal to λ . However, by [Assumption 3\(3\)](#), we have

$$\lambda < \frac{w^e(\mathcal{A}) - w^e(D)}{w^e(\mathcal{A}) - w^e(R)}.$$

This implies that if the elite observe a protest by young citizens, they will never choose to concede democracy. Given this, at the first stage of the game, young citizens in state σ_2 would have an incentive to deviate by choosing NP rather than P , contradicting the assumption that $\mathbb{P}(P|\sigma_2) = 1$.

Let us now verify that $\mathbb{P}(P|\sigma_2) = 0$ constitutes an equilibrium. In this case we must have $\mu(\sigma_1|P) = 1$ which exceeds $\frac{w^e(\mathcal{A}) - w^e(D)}{w^e(\mathcal{A}) - w^e(R)}$. As a result, whenever the elite observe protests, they will always choose to concede democracy. However, under [Assumption 3\(2\)](#), in state σ_2 , young citizens prefer remaining in an autocracy over securing democratic concessions through protests. Therefore, the belief $\mu(\sigma_1|P) = 1$ is consistent with the optimal behavior of young citizens, confirming the equilibrium.

To summarize, in the perfect Bayesian Nash equilibrium, protests occur with probability $\lambda\phi(\psi_y)$, which increases with the youth share. When protests take place, the elite respond by conceding democracy. Conversely, with probability $1 - \lambda\phi(\psi_y)$, no protest occurs, and the economy remains an autocracy.

Appendix B. Additional figures and descriptive evidence

See [Figs. 11–14](#).

Appendix C. LPM regressions with additional controls

Let us briefly discuss the impact of the additional controls — not included in our baseline regressions — on our results. Population size is primarily introduced as a confounder for the youth share, as it appears in the denominator of this variable, and periods of youth population growth are typically also periods of overall population increase. The results reported in columns 2 and 6 of the table reassure us that the effect of the youth share is not driven by population size. The effects of years of schooling attained by the population above 15, urbanization rate, and the Gini coefficient are not statistically significant, whether introduced individually (columns 1, 3, and 4) or when all controls are

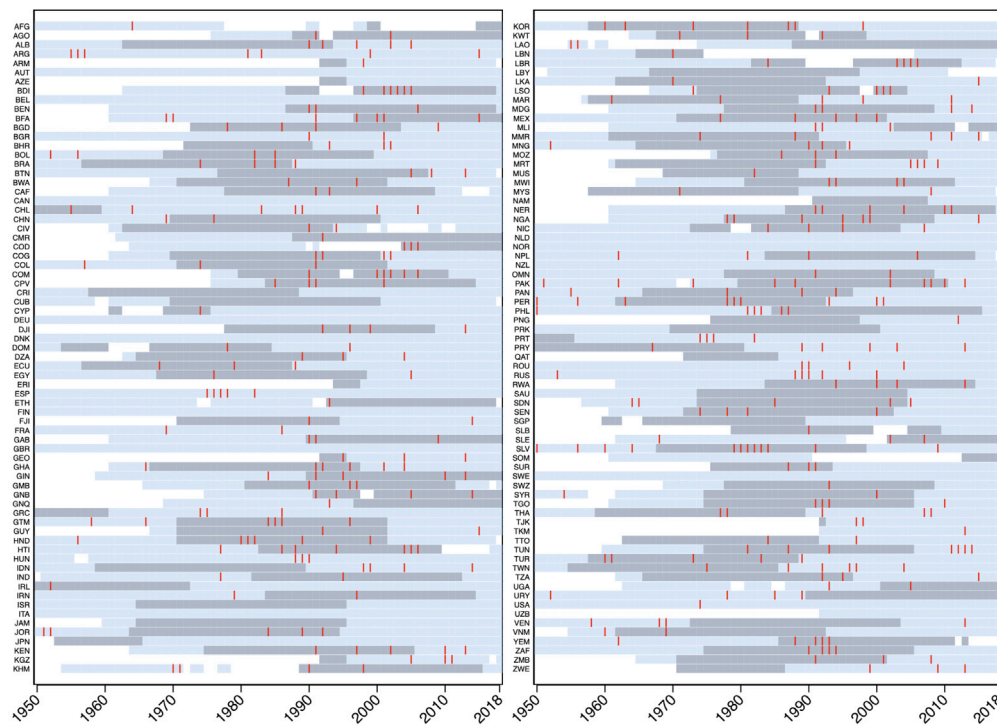


Fig. 14. Demographic transition windows and democratic improvements.

Notes: Each row represents a country and each column a year between 1950 and 2018. Light blue areas indicate years for which data are available but fall outside the demographic transition window. Dark blue areas indicate years within the transition window, defined as a 15-year window centered around the date at which cohorts born at peak fertility reach youth ages. Red vertical lines mark years in which a democratic improvement occurs. White areas correspond to country-years with missing data on democratic improvement. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7

IV estimation with lagged controls.

<i>Panel A: Binary DV: Δ Polity2 ≥ 1</i>				
	IV: Own fertility		IV: Neighbors' fertility	
	(1)	(2)	(3)	(4)
Youth share (15–19) _{<i>t-1</i>}	0.750** (0.34)	0.939** (0.37)	1.679** (0.79)	1.949** (0.89)
Interstate conflicts _{<i>t-16</i>}		-0.035** (0.02)		-0.032* (0.02)
Internal conflicts _{<i>t-16</i>}		0.001 (0.01)		0.000 (0.02)
Urbanization _{<i>t-16</i>}		-0.001 (0.00)		-0.001 (0.00)
<i>Panel B: First stage of 2SLS</i>				
	(1)	(2)	(3)	(4)
Net fertility rate _{<i>t-16</i>}	0.019*** (0.00)	0.019*** (0.00)		
Neighbors' net fertility _{<i>t-21</i>}			0.015*** (0.00)	0.015*** (0.00)
Ceiling control	Yes	Yes	Yes	Yes
Standard controls	Yes	Yes	Yes	Yes
Standard controls _{<i>t-16</i>}	Yes	Yes	Yes	Yes
Country & year FE's	Yes	Yes	Yes	Yes
K-P F-stat on excl. IV's	698.79	625.36	55.51	53.40
Observations	5931	5565	5366	5009
Countries	153	152	151	148

Notes: Robust standard errors clustered by country are in parentheses. Ceiling control equals one if Polity2_{*t-1*} = 10. The standard set of controls includes once-lagged values of the log of GDP per capita, the growth rate of GDP per capita, and the Polity score. Referring to the Kleibergen–Paap (K-P) F-statistic, the test's null hypothesis is that the set of instruments is weak. The panel runs from 1970–2018 for the own net fertility rate specification and from 1975–2018 for the neighborhood net fertility rate specification. ***, **, and * indicate significance at the 0.01, 0.05, and 0.10 levels, respectively.

Table 8
IV estimation neighbors' and additional controls.

Panel A: Binary DV: Δ Polity2 ≥ 1				
IV: Neighbors' fertility				
	(1)	(2)	(3)	(4)
Youth share (15–19) _{t-1}	3.013** (1.20)	3.053** (1.32)	2.519** (1.21)	2.563* (1.32)
Interstate conflicts _{t-1}	0.027 (0.03)	0.027 (0.03)	0.024 (0.03)	0.025 (0.03)
Internal conflicts _{t-1}	-0.038 (0.03)	-0.040 (0.03)	-0.041 (0.03)	-0.042* (0.03)
Urbanization _{t-1}	0.001 (0.00)	0.001 (0.00)	0.001 (0.00)	0.001 (0.00)
Net migration _{t-1}	0.001 (0.00)	0.001 (0.00)	0.000 (0.00)	0.000 (0.00)
Neighbors' ln(GDP p.c.) _{t-1}		-0.006 (0.03)		-0.005 (0.03)
Neighbors' growth rate(GDP p.c.) _{t-1}		-0.003 (0.00)		-0.002 (0.00)
Neighbors' Polity _{t-1}		0.005 (0.00)		0.004 (0.00)
Neighbors' transition _{t-1}			0.240*** (0.08)	0.227*** (0.07)
Panel B: First stage of 2SLS				
	(1)	(2)	(3)	(4)
Neighbors' net fertility _{t-21}	0.018*** (0.00)	0.017*** (0.00)	0.018*** (0.00)	0.017*** (0.00)
Ceiling control	Yes	Yes	Yes	Yes
Standard controls	Yes	Yes	Yes	Yes
Country & year FE's	Yes	Yes	Yes	Yes
K-P F-stat on excl. IV's	38.63	32.97	36.66	31.52
Observations	3316	3316	3296	3296
Countries	133	133	133	133

Notes: Robust standard errors clustered by country are in parentheses. Ceiling control equals one if Polity2_{t-1} = 10. The standard set of controls includes once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score. Referring to the Kleibergen-Paap (K-P) F-statistic, the test's null hypothesis is that the set of instruments is weak. The panel runs from 1970–2018 for the own net fertility rate specification and from 1975–2018 for the neighborhood net fertility rate specification. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

Table 9
Placebo: Reduced form effect of the instruments on urbanization and education.

	OLS				IV – Binary DV: Δ Polity2 ≥ 1		OLS	
	DV: Urbanization		DV: Years of schooling		Instrument: Own fertility	Instrument: Neighbor's fertility	DV: Δ Polity2 ≥ 1	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Net fertility rate _{t-16}	-0.242 (0.52)		-0.179*** (0.06)				0.002 (0.009)	
Neighbors' net fertility _{t-21}		0.070 (0.78)		-0.068 (0.09)				0.017 (0.011)
Youth share (15–19) _{t-1}					0.780** (0.31)	2.102** (0.98)	0.680* (0.38)	0.663** (0.31)
Urbanization rate _{t-1}					-0.001 (0.00)	-0.001 (0.00)		
Years of schooling _{t-1}					0.006 (0.01)	0.007 (0.01)		
Ceiling control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country & year FE's	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
K-P F-stat on excl. IV's	–	–	–	–	626.43	37.40	–	–
Observations	6874	5771	5667	4757	5764	4863	6826	5726
Countries	154	152	132	121	132	130	154	152

Notes: Robust standard errors clustered by country are in parentheses. Ceiling control equals one if Polity2_{t-1} = 10. The standard set of controls includes once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score. Referring to the Kleibergen-Paap (K-P) F-statistic, the test's null hypothesis is that the set of instruments is weak. The panel runs from 1970–2018 for the own net fertility rate specification and from 1975–2018 for the neighborhood net fertility rate specification. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

included (column 6). The share of employment in the manufacturing sector shows a positive and significant impact only when the other controls are excluded (column 5). Moreover, throughout all specifications, the effect of the youth share remains positive and statistically significant. These results, along with the fact that including additional

controls reduces the sample size, lead us to exclude them from our main specifications.

Appendix D. Robustness of the 2SLS estimates

See Tables 7–9.

Table 10
Democratic improvements around demographic transitions.

	Binary DV: $\Delta \text{Polity2} \geq 1$					
	Ordinary least squares					
	(1)	(2)	(3)	(4)	(5)	(6)
Window $[-5, 5]_{t-1}$	0.022** (0.01)					0.020* (0.01)
Window $[-10, 10]_{t-1}$		0.011 (0.01)				
Window $[-15, 15]_{t-1}$			0.012 (0.01)			
Window $[-20, 20]_{t-1}$				0.007 (0.01)		
Window $[-25, 25]_{t-1}$					0.006 (0.01)	
Window $[-25, -15]_{t-1}$						-0.001 (0.01)
Window $[-15, -5]_{t-1}$						0.000 (0.01)
Window $[5, 15]_{t-1}$						0.002 (0.01)
Window $[15, 25]_{t-1}$						-0.008 (0.01)
Ceiling control	Yes	Yes	Yes	Yes	Yes	Yes
Standard controls	Yes	Yes	Yes	Yes	Yes	Yes
Country & year FE's	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,259	10,259	10,259	10,259	10,259	10,259
Countries	132	132	132	132	132	132

Notes: Robust standard errors clustered by country are in parentheses. The unbalanced panel runs from 1850–2018, depending on data availability across countries. Ceiling control equals one if $\text{Polity2}_{t-1} = 10$. The standard set of controls includes once lagged values of: the log of GDP per capita, the growth rate of GDP per capita, and the Polity score. ***/**/* represent significance at the 0.01/0.05/0.10 levels, respectively.

Appendix E. Evidence based on demographic transition timing – results table

See Table 10.

Appendix F. Online appendix

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jdeveco.2026.103875>.

Data availability

Data will be made available on request.

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